
MORTGAGE DELINQUENCY RISK ANALYSIS

Business Case Study

Fannie Mae Single-Family Loan Performance Dataset | 2020 Q1-Q2

Prepared by	Chegwe Favour
Role	Credit Risk & Data Analyst
Date	2026
Dataset	Fannie Mae SF Loan Performance — 2020 Q1-Q2
Portfolio Scope	84.18M Servicing Observations 1.92M Unique Loans (Snapshot)
Tools Used	PostgreSQL Power BI Excel Python
Portfolio	https://chegwefavourportfoliioo.netlify.app/
GitHub	https://github.com/favouritefavil
LinkedIn	http://www.linkedin.com/in/favour-chegwe

This case study documents the complete analytical journey of a portfolio-grade mortgage credit risk analysis project — from business problem definition through SQL analysis, feature engineering, Power BI dashboard development, and executive reporting.

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1 EXECUTIVE SUMMARY

This case study documents the end-to-end development of a portfolio-grade mortgage delinquency risk analysis project built on the Fannie Mae Single-Family Loan Performance Dataset for 2020 Q1 and Q2. The project was designed to simulate the kind of credit risk analytics work performed by Risk Analytics teams within banks, mortgage institutions, and financial services firms.

The business problem was straightforward but significant: given a large mortgage portfolio, which borrowers are most likely to become delinquent, which structural and geographic characteristics amplify that risk, and how should management prioritise monitoring resources to prevent credit losses from materialising.

The analysis was conducted on 84.18 million monthly loan-performance observations using PostgreSQL for analytical layer development and feature engineering. For executive reporting and dashboard development, the dataset was transformed into a latest-snapshot fact table of approximately 1.92 million unique loans — a deliberate data modelling decision that ensures portfolio KPIs reflect current loan status rather than cumulative historical balances. The significance of this architectural choice is explained in detail in Section 7.

The Power BI dashboard was structured across six pages, each answering one distinct executive business question: portfolio health, borrower risk, loan and property structure, geographic risk diagnosis, behavioural risk patterns, and portfolio surveillance. The dashboard was designed for a Chief Risk Officer audience, with each page building naturally on the previous to create a coherent analytical narrative.

Figure 4. Project Lifecycle

End-to-end workflow from business understanding to portfolio publication and career preparation.

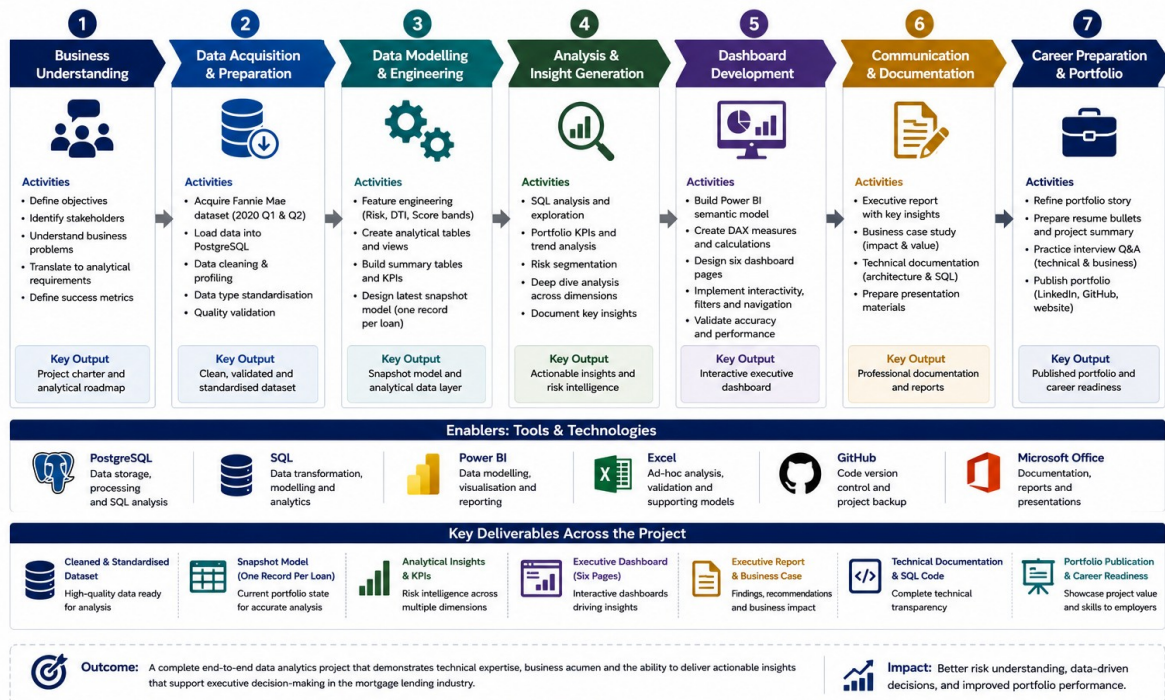


Figure 1. Project Lifecycle

End-to-end workflow illustrating the seven-stage project methodology — from Business Understanding through Career Preparation — showing all activities, tools, deliverables, and outcomes.

Six-page Power BI dashboard showing the complete navigation structure. Each page answers one distinct executive business question, progressing from portfolio health through to operational surveillance.

Key Findings

- The portfolio delinquency rate stands at 0.65% in the snapshot model, with 99.3% of loans performing as contractually expected.
- Modified loans — representing less than 1% of portfolio exposure — exhibit a delinquency rate of 22.8%, approximately 38 times higher than non-modified loans.
- Borrower credit quality is the strongest individual predictor of delinquency, with Very High Risk borrowers exhibiting rates more than 22 times those of Very Low Risk borrowers.
- Risk layering — multiple concurrent risk factors — amplifies delinquency non-linearly, with High Risk segment delinquency at 2.4% versus 0.3% for Low Risk borrowers.
- New York exhibits the most concerning geographic risk profile, combining the highest delinquency rate (1.18%) with the highest modification concentration (1.03%) across all major markets.
- California, despite representing approximately 16.6% of portfolio exposure, demonstrates strong underlying credit quality with a delinquency rate of only 0.47%.
- The portfolio surveillance framework identifies Modified Loans as the only segment currently in breach of its defined monitoring threshold.

KEY INSIGHT *A single-metric view of portfolio health — delinquency rate alone — would suggest a well-performing portfolio. The analytical value of this project lies in the layers beneath that headline: which segments are driving risk, why, and how much financial exposure is concentrated in those segments.*

Project value: *This project demonstrates the ability to translate 84 million rows of raw institutional data into actionable executive insights using industry-standard credit risk frameworks, professional-grade SQL engineering, and executive dashboard design — all attributes directly applicable to Credit Risk Analyst and Finance Analytics roles.*

2 BUSINESS BACKGROUND

Mortgage Lending

A mortgage loan is a form of secured credit extended to individuals for the purpose of purchasing or refinancing residential property. The property itself serves as collateral, giving the lender a legal claim on the asset in the event that the borrower fails to meet payment obligations. Mortgages are typically long-term instruments — spanning 15 to 30 years — and represent the single largest debt obligation in most borrowers' financial lives.

In the United States, the mortgage market is supported by Government-Sponsored Enterprises (GSEs), including Fannie Mae and Freddie Mac, which purchase conforming mortgages from originating lenders, package them into mortgage-backed securities (MBS), and provide guarantees that support the secondary market. This process allows banks and mortgage companies to originate new loans continuously by recycling capital from prior lending activity.

Banks generate revenue from mortgages through interest income over the loan term, origination fees charged at closing, servicing fees for collecting payments and managing the loan, and gains on sale when loans are sold into the secondary market at premium prices.

Mortgage Delinquency

Mortgage delinquency occurs when a borrower fails to make a scheduled payment by its due date. Most mortgage contracts include a 15-day grace period before a payment is classified as late. Delinquency is typically first reported at 30 days past due and escalates in 30-day increments — 30 DPD, 60 DPD, 90 DPD — reflecting increasing severity of borrower financial distress.

The transition from delinquency to default — formally defined as 90 or more days past due under Basel credit risk frameworks and IFRS 9 — is the threshold at which lenders are required to significantly increase loan loss provisions and begin active legal loss mitigation processes, including loan modification, forbearance negotiation, or ultimately foreclosure.

Delinquency Stage	Days Past Due	Business Status	Risk Level
Current	0–14	Performing	Low
30 DPD	30–59	Early delinquency	Moderate
60 DPD	60–89	Mid delinquency	High
90+ DPD	90+	Serious delinquency	Critical
Foreclosure	N/A	Legal action initiated	Extreme
REO	N/A	Bank-owned post-foreclosure	Loss realised

CREDIT RISK PERSPECTIVE Under IFRS 9, a loan reaching 90+ DPD triggers Stage 3 classification — requiring the bank to recognise lifetime Expected Credit Loss (ECL) rather than 12-month ECL. This shift in provisioning requirement has immediate capital and profitability consequences, which is why portfolio managers monitor 90+ DPD rates as a primary credit health indicator.

Why Delinquency Matters

Delinquency is the single most important early warning indicator in mortgage portfolio management. A loan that misses one payment is not necessarily a loss — the majority of 30 DPD borrowers cure (return to current status) within one or two payment cycles without formal intervention. However, a loan that reaches 90+ DPD is statistically unlikely to cure without active modification or workout, and represents an exposure that will likely result in some degree of credit loss.

From an institutional perspective, rising delinquency rates have cascading financial consequences. They interrupt expected interest income streams. They trigger increases in loan loss provisions under IFRS 9 and Basel frameworks (Expected Loss = Probability of Default × Exposure at Default × Loss Given Default). They consume regulatory capital. They attract scrutiny from regulators and investors. And in extreme cases, they can threaten the solvency of heavily leveraged lending institutions — as demonstrated during the 2008 global financial crisis.

EXECUTIVE TAKEAWAY *Every 1% increase in the portfolio delinquency rate on a \$194bn portfolio moves approximately \$1.94bn into delinquent status. At an assumed LGD of 30%, that represents approximately \$582M in additional expected credit loss. This is why monitoring delinquency trends — not just their current level — is a continuous management imperative.*

Why Lenders Monitor Portfolio Risk

Mortgage portfolio monitoring is the operational practice by which a bank or servicer tracks the ongoing performance of its loan book, identifies deteriorating segments before they become losses, and allocates monitoring and collections resources efficiently. An institution that can identify — with precision — which borrowers are most likely to default in the next 30 to 90 days can intervene early, offer forbearance or modification options, and in many cases prevent a delinquency from progressing to foreclosure. Early intervention is almost always cheaper than late-stage loss mitigation.

Effective portfolio monitoring requires both breadth and depth. Breadth — understanding the overall portfolio's health through headline KPIs. Depth — understanding which specific segments, geographies, or borrower profiles are driving risk, and why. This project was designed to deliver both.

PORTFOLIO MANAGEMENT NOTE *Best-practice mortgage portfolio monitoring incorporates at least five dimensions: borrower credit quality trends, loan structure risk, collateral value monitoring (HPI tracking), geographic concentration assessment, and behavioural indicators (modification rates, payment history). This project addresses all five dimensions across its six-page dashboard.*

The 2020 COVID-19 Context

The reporting period for this analysis — January through June 2020 — coincides with the onset of the COVID-19 pandemic. The United States mortgage market experienced an extraordinary policy response during this window: millions of borrowers were offered payment forbearance under the CARES Act, foreclosure moratoria were implemented preventing lenders from initiating or continuing foreclosure proceedings, and payment deferral options allowed borrowers to move missed payments to the end of their loan term without immediate delinquency classification.

These interventions created a critical analytical nuance. Reported delinquency rates during 2020 Q1–Q2 significantly understate the true extent of borrower financial stress, because borrowers in active forbearance were classified as current under the terms of their relief agreements. Foreclosure activity during this period was effectively zero — not because borrowers were uniformly healthy, but because legal proceedings were suspended.

Analytical implication: *The portfolio metrics in this project represent a floor of credit stress rather than its ceiling. The true performance of the portfolio became visible only in subsequent quarters as forbearance programmes expired. This context was explicitly documented throughout the project and shapes the interpretation of all reported metrics.*

3 BUSINESS PROBLEM

The fundamental business problem this project addresses is one that every mortgage-holding institution faces continuously: given a large, heterogeneous loan portfolio, how does management distinguish the loans most likely to become delinquent from those that will continue to perform?

This distinction matters because mortgage portfolios are not uniform. Borrowers differ in credit quality, income stability, leverage position, and financial resilience. Loans differ in structure, purpose, and collateral quality. Geographies differ in economic conditions and housing market dynamics. When a lender treats all loans identically — applying the same monitoring intensity, the same collections outreach, the same modification criteria — it wastes resources on low-risk performing loans while under-serving the high-risk segments where intervention would create the most value.

From the perspective of a Credit Risk Manager, the core question is not simply 'how many loans are delinquent?' — that can be answered with a single KPI. The deeper, more valuable question is: 'Which specific borrower, loan, property, and geographic characteristics are most strongly associated with delinquency, and how do those characteristics interact to compound risk?' Answering that question requires structured, multi-dimensional analysis of the full loan portfolio.

This project was designed to answer that deeper question systematically — starting from raw data, building an analytical layer through SQL engineering, extracting risk driver insights through exploratory data analysis, and communicating findings through a structured executive dashboard that enables decision-makers to see, understand, and act on portfolio risk.

Business problem statement: *A mortgage portfolio of 1.92 million loans requires a structured analytical framework to identify the borrower, structural, geographic, and behavioural characteristics most strongly associated with delinquency, quantify the financial exposure within each risk segment, and translate those findings into an operational monitoring framework that management can use to prevent credit losses from materialising.*

4 BUSINESS OBJECTIVES

The project was structured around six interconnected business objectives, each mapping to a specific dimension of mortgage portfolio risk analysis:

Objective 1 — Establish Portfolio Health Baseline

Before any risk driver analysis can be meaningful, management needs a clear, accurate picture of overall portfolio health. This objective required calculating and presenting portfolio-level KPIs — delinquency rate, serious delinquency rate, current performing rate, total exposure, and delinquent exposure — in a format immediately accessible to a Chief Risk Officer.

Objective 2 — Identify Borrower-Level Risk Drivers

The project aimed to determine which borrower characteristics — credit score, debt-to-income ratio, and risk segment — are most strongly and consistently associated with mortgage delinquency. This required segmenting the portfolio by each variable, calculating delinquency rates within each segment, and identifying both the magnitude and ordering of risk factors.

Objective 3 — Assess Structural and Collateral Risk

Loan and property characteristics create structural risk independent of borrower quality. This objective required evaluating the delinquency implications of loan purpose, loan-to-value ratio, property type, and amortisation type — identifying which structural factors materially amplify portfolio risk.

Objective 4 — Diagnose Geographic Risk Concentration

Geographic concentration creates systemic exposure that individual loan-level analysis cannot capture. This objective required not only identifying which states carry the most exposure and the highest delinquency rates, but diagnosing why — determining whether geographic risk reflects local economic conditions or simply reflects the composition of borrowers in those markets.

Objective 5 — Analyse Behavioural Distress Patterns

Loan modification status is a powerful behavioural signal that goes beyond origination characteristics. This objective required isolating the delinquency impact of modification status — determining whether it operates independently of existing risk segmentation, and quantifying the disproportionate concentration of delinquent exposure within the modified loan segment.

Objective 6 — Build an Operational Surveillance Framework

Analysis without action structure has limited management value. The final objective was to translate the risk findings from Objectives 1–5 into a formal portfolio surveillance framework: defined monitoring segments, quantified risk thresholds, escalation status indicators, and a priority ranking system that management can use to allocate oversight resources efficiently.

5 EXECUTIVE BUSINESS QUESTIONS

The following business questions guided the analytical approach and the structure of the Power BI dashboard. Each question was selected because it represents a genuine management decision point — not merely an analytical curiosity.

#	Business Question	Dashboard Page	Analytical Value
1	How healthy is the overall mortgage portfolio today?	Page 1 — Executive Overview	Establishes portfolio health baseline for executive orientation
2	Which borrowers are most likely to become delinquent?	Page 2 — Borrower Risk	Identifies the primary human risk factors driving delinquency
3	Which loan and property structures carry elevated risk?	Page 3 — Property & Loan	Separates structural risk from borrower risk for policy targeting
4	Why are certain geographic markets riskier than others?	Page 4 — Geographic Risk	Diagnoses whether geographic risk reflects location or composition
5	How does modification status influence delinquency severity?	Page 5 — Behavioural Risk	Establishes modification as independent behavioural risk signal
6	Which portfolio segments require continuous management attention?	Page 6 — Portfolio Surveillance	Converts findings into an operational monitoring framework
7	Which risk factors compound delinquency when combined?	Pages 2 and 4	Supports risk-layering policy and multi-factor underwriting
8	Where is delinquent exposure financially concentrated?	Pages 2, 3, and 5	Ensures monitoring is driven by exposure, not just loan counts
9	What actions should management take based on these findings?	Page 6 and Exec Report	Translates analysis into concrete risk management recommendations

6 DATASET OVERVIEW

Source and Coverage

The dataset used in this project is the Fannie Mae Single-Family Loan Performance Dataset, one of the most comprehensive publicly available sources of US mortgage performance data. Fannie Mae releases this data quarterly as part of its commitment to transparency in the mortgage-backed securities market.

The dataset covers single-family residential mortgage loans acquired by Fannie Mae through its conforming conventional loan programmes. It includes origination characteristics captured at loan inception and monthly performance records tracking each loan's evolving status through its lifecycle.

Dataset Source	Fannie Mae Single-Family Loan Performance Dataset
Reporting Period	2020 Q1 (January–March) and 2020 Q2 (April–June)
Total Records	84,223,406 raw records (before cleaning)
After Cleaning	84,183,216 records (40,190 null credit score records removed)
Snapshot Model	1,920,000 unique loans (one record per loan, latest observation)
Fields	108 columns per record (Column1–Column108)
File Format	Pipe-delimited text files, no header row
Grain	Loan × Monthly Reporting Period (loan-month panel)

The Loan-Month Panel Structure

A critical aspect of this dataset that shapes the entire analytical approach is its grain: each row represents one loan in one calendar month. A loan originated in January 2020 and still active in June 2020 would appear in six separate rows — one for each monthly reporting period. This means the raw dataset does not contain 84 million unique loans; it contains 84 million monthly observations across a much smaller number of unique loans.

This structure is appropriate for historical analysis — tracking how individual loans evolve through delinquency stages over time — but it creates a double-counting problem for current-state portfolio reporting. A loan that was delinquent in March 2020 but cured by June 2020 would contribute two different delinquency status records to any aggregate count. For this reason, a distinct reporting model was developed for the Power BI dashboard, described in detail in Section 7.

TECHNICAL NOTE The dataset contains no column header row. All 108 fields are identified by pipe-delimited position. A complete field-mapping exercise was performed, translating Column1 through Column108 to official Fannie Mae glossary names, business definitions, and appropriate data types before any analytical work began.

Key Variables

Variable	Source Column	Business Role	Category
Loan Identifier	Column 2	Primary key — links all monthly records to one loan	Identifier
Monthly Reporting Period	Column 3	Time dimension — MMYYYY format	Identifier
Borrower Credit Score	Column 24	Primary PD predictor — strongest delinquency signal	Borrower Risk
DTI Ratio	Column 23	Payment capacity — repayment stress	Borrower Risk

Variable	Source Column	Business Role	Category
		indicator	
Number of Borrowers	Column 22	Income stream count — risk mitigation factor	Borrower Risk
Original LTV	Column 20	Collateral coverage — primary LGD driver	Loan Structure
Original CLTV	Column 21	Combined leverage including junior liens	Loan Structure
Original Interest Rate	Column 8	Payment burden and risk-based pricing proxy	Loan Structure
Original Loan Term	Column 13	Exposure duration — 30yr vs 15yr risk profile	Loan Structure
Loan Purpose	Column 27	Purchase vs refinance vs cash-out risk distinction	Loan Structure
Amortization Type	Column 35	FRM vs ARM — payment shock risk identification	Loan Structure
Property Type	Column 28	Collateral type — liquidity and value stability	Collateral Risk
Occupancy Status	Column 30	Owner vs investor — payment priority signal	Collateral Risk
Property State	Column 31	Geographic risk — foreclosure law, HPI, economy	Geographic
Current Delinquency Status	Column 40	Target variable — DPD stage in current month	Performance
Loan Payment History	Column 41	24-month behavioural string — strongest behavioral signal	Performance
Modification Flag	Column 42	Distress event marker — modified loan identification	Performance
Zero Balance Code	Column 44	Terminal outcome — credit loss vs healthy exit	Performance
Current Actual UPB	Column 12	Current exposure — EAD input	Performance

7 DATA ARCHITECTURE

Design Philosophy

The data architecture for this project was designed around a single governing principle: analytical integrity at every layer. Every transformation — from raw import through feature engineering to dashboard reporting — was documented, justified, and reproducible. This approach mirrors the standards expected in a professional credit risk analytics environment, where data lineage, methodology transparency, and audit trails are non-negotiable requirements.

Source Dataset Architecture

The source dataset is a single integrated mortgage performance file. Origination characteristics — credit score, LTV, DTI, loan purpose, property type — are repeated on every monthly row for each loan. Performance characteristics — delinquency status, current UPB, modification flag — update each month. This integrated structure means that separating origination attributes from performance data requires deliberate query design rather than separate source files.

Architecture decision: *The source dataset is a single integrated mortgage performance dataset. Origination characteristics are repeated across monthly performance records. For analytical purposes, the project separates the data into logical origination and performance layers within PostgreSQL rather than relying on separate source files.*

Figure 1. Solution Architecture

End-to-end analytical workflow from raw mortgage servicing data to executive decision support.

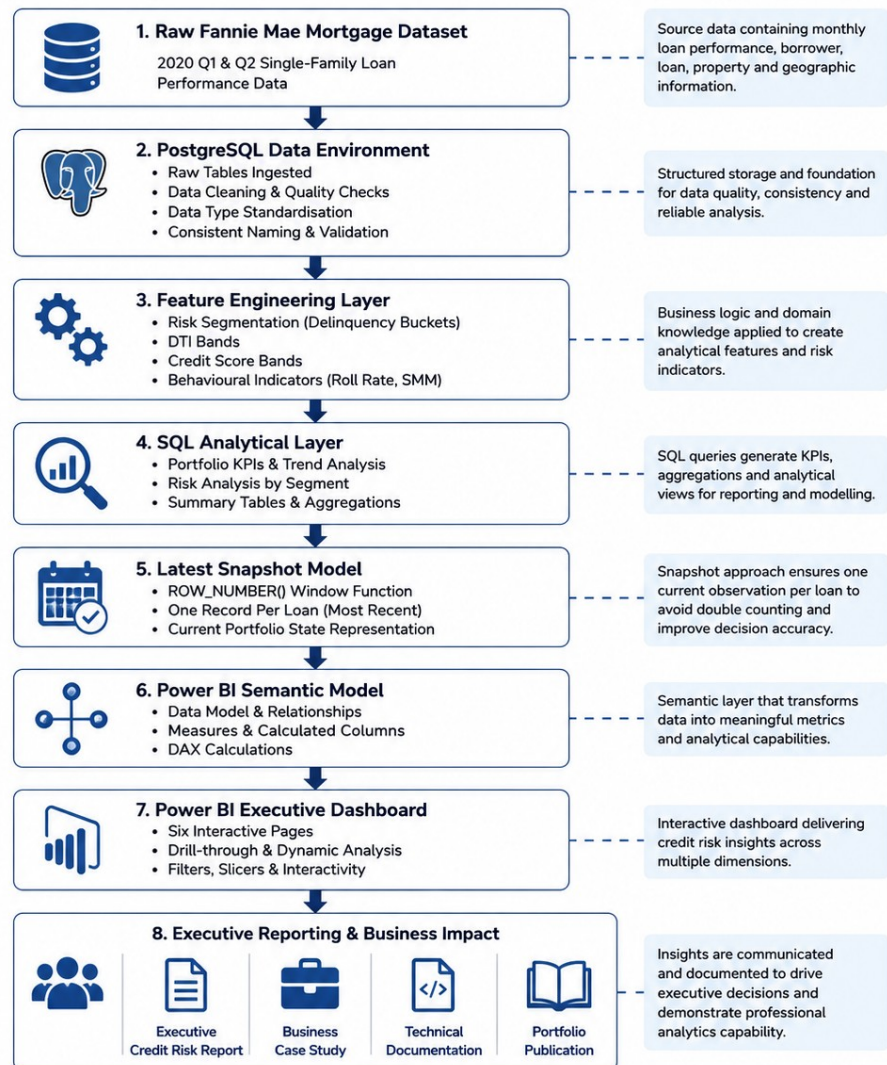


Figure 1. End-to-end analytical architecture illustrating the transformation of raw Fannie Mae mortgage servicing data into executive credit risk intelligence through PostgreSQL, feature engineering, snapshot modelling, Power BI reporting, and professional documentation.

Figure 2. Solution Architecture

Eight-layer analytical architecture showing the transformation of raw Fannie Mae mortgage servicing data into executive credit risk intelligence through PostgreSQL, feature engineering, snapshot modelling, Power BI reporting, and professional documentation.

Four-layer data pipeline showing flow from raw pipe-delimited files through cleaned view, typed view, and feature-engineered view to the pre-aggregated summary tables and latest-snapshot fact table used in Power BI.

PostgreSQL Analytical Layer

All data engineering, transformation, and analysis was performed in PostgreSQL. The analytical architecture was built across four sequential layers:

Layer	Object Name	Record Count	Purpose
1 — Raw Import	mortgage_raw_q1 / mortgage_raw_q2	84.2M rows	Raw pipe-delimited data, all columns as TEXT
2 — Cleaned View	mortgage_clean_v	84.18M rows	Null credit score records removed (40,190 records)
3 — Typed View	mortgage_typed_v	84.18M rows	Numeric and categorical fields cast to correct data types
4 — Feature Engineered	mortgage_fe_v	84.18M rows	Derived risk features added — primary analytical source

SQL summary tables were created from mortgage_fe_v for use in exploratory data analysis, KPI calculation, and segmentation work. These pre-aggregated tables significantly improved query performance across the 84 million row dataset and enabled the iterative analytical workflow required for EDA.

The Latest-Snapshot Reporting Model — A Defining Architectural Decision

The most significant architectural decision in this project — and one that demonstrates genuine data modelling judgement — was the creation of a separate fact table for Power BI reporting. Understanding why this decision was necessary requires understanding the difference between what the data shows and what executives need to see.

The analytical panel (84.18 million rows) records one observation per loan per calendar month. This is the correct grain for historical analysis: it enables roll rate analysis, cure rate tracking, vintage performance curves, and time-series delinquency monitoring. But it is fundamentally the wrong grain for current-state executive KPIs, and using it for executive reporting without transformation would produce systematically misleading results.

Consider the core problem. If Total Portfolio Exposure were calculated across all 84.18 million monthly rows, the result would be the sum of each loan's outstanding balance across every month it appeared in the dataset. A loan that has been in the portfolio for 24 months would have its balance counted 24 times. The result — \$19.97 trillion — sounds like a staggeringly large mortgage portfolio, because it is a cumulative historical figure, not a current portfolio balance. No institution could credibly report \$19.97 trillion as its current mortgage exposure.

The same problem affects every current-state metric. A delinquency rate calculated across the full panel will include loans that were delinquent at some point in their history but have since cured — overstating the current delinquency level. A loan count from the panel would count each loan multiple times — overstating the portfolio population.

Figure 2. Snapshot Model Architecture (Latest Snapshot Approach)

One record per loan representing the most recent state to enable accurate portfolio-level risk analysis.

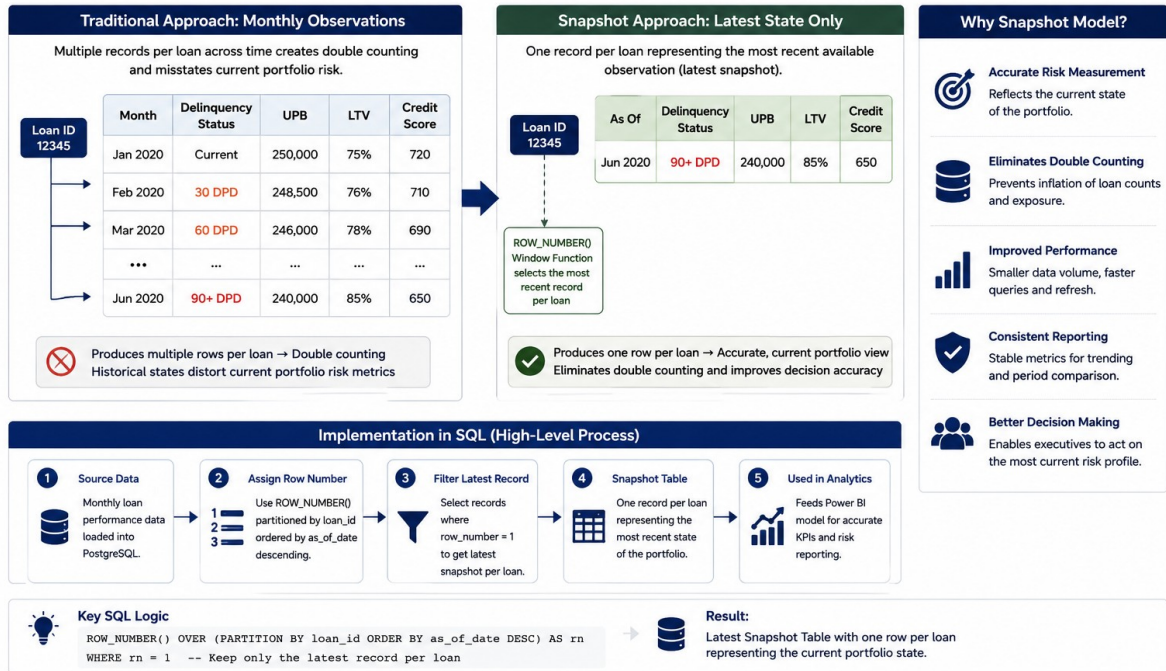


Figure 3. Snapshot Model Architecture

Comparison of the traditional monthly observation approach (left) and the latest-snapshot model (right), illustrating how the ROW_NUMBER() window function selects one record per loan to eliminate double counting and produce accurate current-state portfolio KPIs.

Diagram illustrating the grain problem: the 84.18M observation panel versus the 1.92M unique loan snapshot. Left side shows how loan balances accumulate across monthly rows. Right side shows the snapshot model selecting only the most recent observation per loan.

To solve this, a latest-snapshot fact table was created by selecting the most recent monthly observation for each unique loan. This produces approximately 1.92 million unique loan records, each representing the current state of that loan at its most recent reporting date. Executive KPIs calculated from this table accurately reflect what a CRO actually needs to know: how many loans are currently in the portfolio, what is their current delinquency status, and how much exposure is at risk today.

KEY INSIGHT The 84.18M panel answers: 'What has happened historically across the portfolio's full servicing lifecycle?' The 1.92M snapshot answers: 'What does the portfolio look like today?' Both questions are valid — but they require different data grains. Using the wrong grain for either question produces analytically incorrect results.

This decision introduced a secondary effect that required explicit documentation: the snapshot model produces lower delinquency rates than the panel analysis. This is not a data quality issue — it is an expected and correct consequence of the different grains. The snapshot captures each loan's current status; the panel averages across all historical statuses including historically delinquent loans that have since cured. Both figures are correct in their respective contexts. The reporting note embedded in the Executive Credit Risk Report and the dashboard header explicitly communicates this distinction to prevent misinterpretation.

Data modelling judgement: The decision to build a separate latest-snapshot table — rather than applying filters to the panel — reflects an understanding that data grain is not a technical detail but a business question: what time horizon should executive metrics represent? For current-state portfolio reporting, the answer is always the most recent available position, not cumulative history.

Field Mapping and Column Naming

The raw Fannie Mae dataset contains no column headers. Fields are identified by their position in the pipe-delimited file. A complete field mapping exercise was performed, translating all 108 columns (Column1 through Column108, where Column1 is a blank leading-pipe artifact) to their official Fannie Mae glossary names, business definitions, appropriate data types, analytical relevance, and priority classification.

Columns were classified as Critical (36 fields — core risk drivers, target variable, KPI inputs), Useful (22 fields — enrichment variables for segmentation), Low Priority (15 fields — operational or supplementary), and Ignore (35 fields — CAS/CIRT-only fields not applicable to SF Loan Performance analysis). This classification governed which fields were carried into the analytical layer.

8 DATA CLEANING AND FEATURE ENGINEERING

Data Profiling

Before any cleaning or transformation, a comprehensive data profiling exercise was conducted across all key analytical variables. The objective was to evaluate missing value rates, validate ranges against official Fannie Mae documentation and industry underwriting standards, confirm categorical enumerations, and identify any data quality issues that could corrupt downstream analysis.

Variable	Min	Max	Missing Values	Validation Outcome
Credit Score	620	825	40,190	Values within expected mortgage underwriting range
LTV Ratio	11	97	0	Values within Fannie Mae conforming limits
CLTV Ratio	12	102	0	Values within acceptable Fannie Mae range
DTI Ratio	1	50	0	Values within expected underwriting range
Modification Flag	Y/N	Y/N	995,640	Null records treated as non-modified — expected
Delinquency Status	00	70 / XX	0	Values consistent with performance reporting standards
Zero Balance Code	01-96	Various	83.2M	Majority null — expected for active performing loans

KEY INSIGHT The dataset demonstrated exceptionally strong data quality across 84+ million records. Credit Score was the only variable requiring record removal. All core underwriting variables — LTV, CLTV, DTI — were fully populated and within valid business ranges. This is a reflection of the rigorous data standards applied by Fannie Mae to its acquired loan portfolio.

Cleaning Decision

One cleaning decision was made: records with missing credit score were removed. Credit score is the primary borrower risk variable used throughout the segmentation and delinquency analysis. Without a credit score, a loan cannot be reliably classified into any risk segment, and its inclusion would create noise in the most analytically important dimension of the project.

The 40,190 records removed represented 0.048% of the total dataset — a negligible removal rate that had no material impact on portfolio representation. No imputation was performed. The conservative approach of removal rather than imputation was appropriate given the critical role of credit score in the analytical framework and the minimal volume of affected records.

All other variables — including the additional Zero Balance Codes identified in the 2020 dataset that were not documented in the glossary version reviewed — were retained. These additional codes reflect dataset version differences rather than data quality issues, and their presence was documented as an analytical observation.

Data Type Standardisation

The raw PostgreSQL import loaded all 108 columns as TEXT to ensure successful ingestion of the large files. A typed view (mortgage_typed_v) was subsequently created to cast each analytical field to its appropriate data type. This standardisation was essential for reliable numeric calculations, efficient aggregation, and correct comparison logic in all downstream SQL queries.

Key data type decisions: Delinquency Status, Zero Balance Code, MSA code, and Zip Code were retained as TEXT despite containing numeric-looking values. Casting these to INTEGER would destroy leading zeros and break all filtering and grouping logic. Date fields in MMYYYY format were also retained as TEXT because this format is not a valid ISO date — conversion to PostgreSQL DATE type would require a transformation function and was only applied where date arithmetic was genuinely needed.

TECHNICAL NOTE Nullable numeric fields used the pattern: `NULLIF(column, '')::NUMERIC` rather than a plain cast. Raw imports from pipe-delimited files frequently store missing values as empty strings rather than SQL NULL. A plain cast would throw an error on empty strings; the `NULLIF` conversion cleanly handles them without data loss.

Feature Engineering

Feature engineering transformed raw mortgage attributes into business-focused risk indicators aligned with underwriting standards and credit risk management practice. The following features were created:

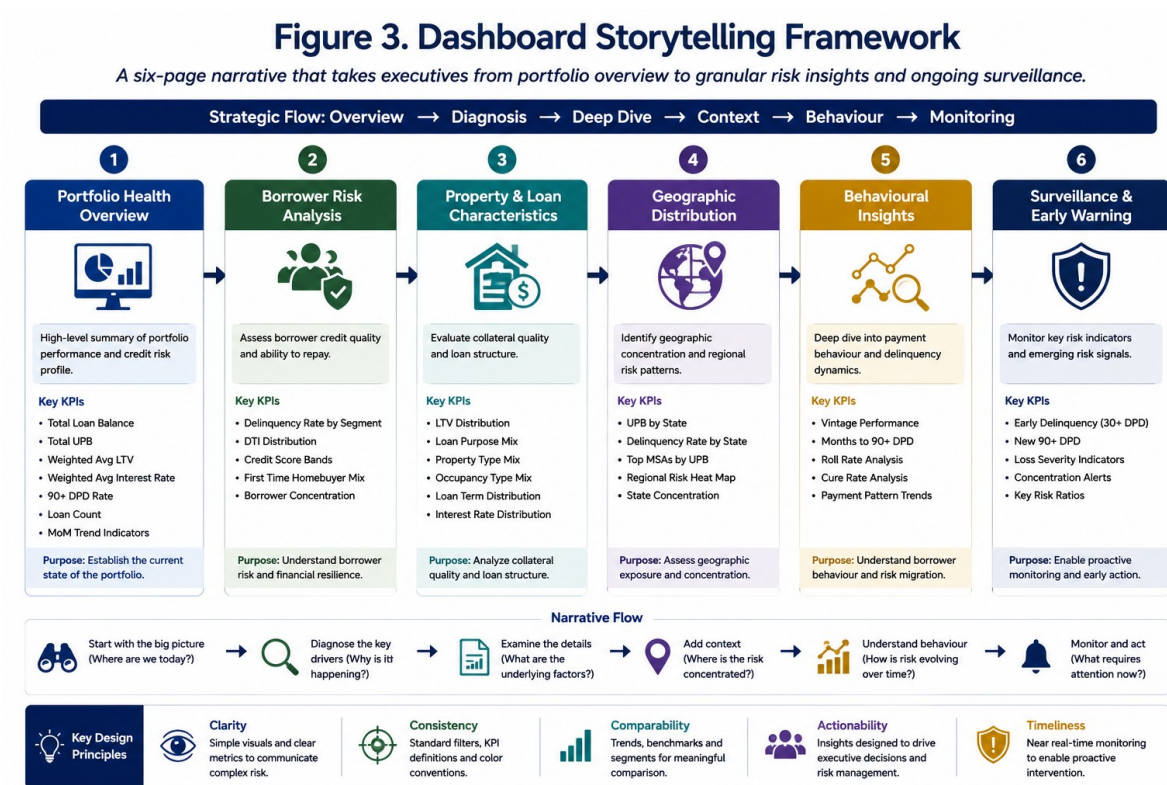


Figure 4. Dashboard Storytelling Framework

Six-page narrative framework showing the strategic flow from Portfolio Health Overview through Borrower Risk, Property & Loan Characteristics, Geographic Distribution, Behavioural Insights, and Surveillance & Early Warning, with the key KPIs and design principles governing each page.

Flow diagram showing the transformation from raw variables (Credit Score, LTV, DTI) through derived features (Credit Score Band, LTV Band, DTI Band) to composite risk indicators (Risk Factor Count, Risk Segment, Delinquency Flag, Serious Delinquency Flag).

Feature	Logic	Thresholds	Business Rationale
Credit Score Band	CASE WHEN credit_score classification	Very Low Risk (≥ 780), Low (740–779), Moderate (680–739), High	Segments borrowers by credit quality tier consistent with

Feature	Logic	Thresholds	Business Rationale
		(620–679), Very High Risk (<620)	mortgage industry underwriting standards
LTV Band	CASE WHEN orig_ltv classification	Very Low (<60%), Low (60–80%), Moderate (80–90%), High (>90%)	Segments loans by collateral coverage and PMI threshold — key LGD driver
DTI Band	CASE WHEN dti_ratio classification	Low (<28%), Moderate (28–36%), High (36–43%), Very High (>43%)	Segments by payment capacity against industry guidelines (28/36 rule, 43% Fannie Mae max)
Risk Factor Count	SUM of risk conditions present	0–3 points: Credit Score <680 (+1), LTV >90 (+1), DTI >43 (+1)	Measures cumulative risk exposure — foundation for risk-layering analysis
Risk Segment	CASE WHEN risk_factor_count	Low Risk (0), Moderate Risk (1), High Risk (2–3)	Creates the primary multi-variable risk classification used throughout dashboard and EDA
Delinquency Flag	delinquency_status != '00'	Binary: 1 = delinquent, 0 = current	Target variable for delinquency rate calculations across all segments
Serious Delinquency Flag	delinquency_status >= '03'	Binary: 1 = 90+ DPD, 0 = otherwise	Identifies loans at the Basel/IFRS 9 formal default threshold

Feature engineering philosophy: Every derived feature was designed to answer a specific credit risk business question, not simply to create additional columns. Each threshold was chosen based on industry standards — Fannie Mae guidelines, Basel default definition, IFRS 9 staging thresholds — rather than statistical optimisation alone. This grounds the analysis in the language and frameworks that credit risk professionals actually use.

Following feature engineering, the primary analytical view (mortgage_fe_v) was used to create pre-aggregated summary tables for EDA and KPI development. These summary tables pre-calculated segment-level metrics — loan counts, exposure totals, delinquency rates — across the key analytical dimensions, enabling rapid iterative querying without repeatedly scanning 84 million rows.

9 ANALYTICAL METHODOLOGY

Analytical Framework

The analytical methodology was structured around a progressive risk decomposition framework: beginning with portfolio-level health metrics, then drilling down systematically through borrower, structural, geographic, and behavioural risk dimensions. This approach mirrors the way credit risk analysis is actually conducted in financial institutions — starting with the aggregate picture and then interrogating the components.

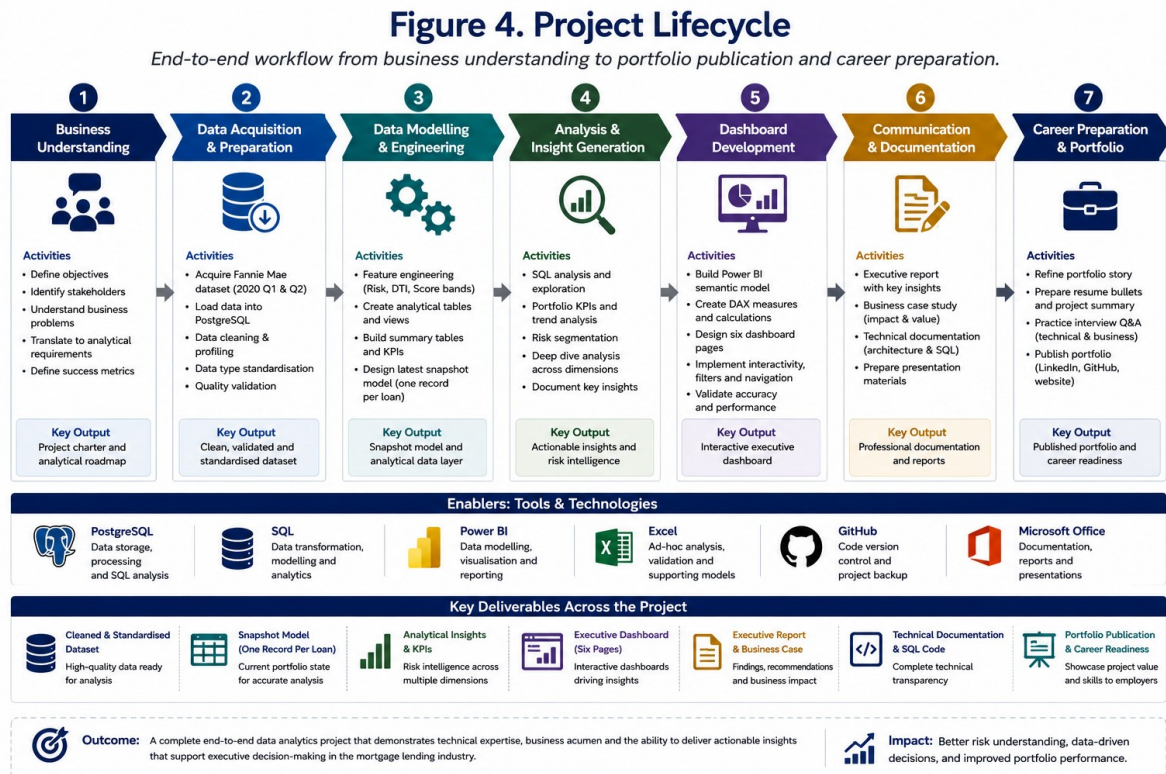


Figure 5. Project Lifecycle — Development Stages

Detailed breakdown of all seven project stages, enabler tools and technologies (PostgreSQL, SQL, Power BI, Excel, GitHub, Microsoft Office), and the full set of key deliverables produced across the project lifecycle.

Pyramid diagram showing five analytical layers: Portfolio Health (base) → Borrower Risk → Structural Risk → Geographic Risk → Behavioural Risk (apex). Each layer adds explanatory depth to the findings of the layer below.

Portfolio KPI Development

The first analytical step was establishing the portfolio health baseline. The following KPIs were calculated from the SQL analytical layer across the full 84.18 million observation panel, alongside their corresponding dashboard snapshot values:

KPI	SQL Panel Value	Dashboard Snapshot	Interpretation of Difference
Delinquency Rate	1.43%	0.65%	Panel includes historical DQ including cured loans. Snapshot shows current DQ status only.

KPI	SQL Panel Value	Dashboard Snapshot	Interpretation of Difference
Serious Delinquency Rate	0.66%	0.22%	Same grain difference applies. Snapshot reflects current 90+ DPD position.
Portfolio Exposure	\$19.97 Trillion	\$194 billion	Panel sums monthly UPB across all observations. Snapshot is current outstanding balance of unique loans.
Delinquent Exposure	\$326.55 Billion	\$2.6 billion	Panel accumulates delinquent UPB across all months. Snapshot is current delinquent balance.
Current Loan Rate	98.56%	99.3%	Snapshot captures most recent status — higher current rate as some historical DQ loans have cured.

EXECUTIVE TAKEAWAY *The differences between SQL panel values and dashboard snapshot values are not discrepancies — they are expected and correct consequences of two different analytical grains. The panel values were used for historical analysis and EDA. The snapshot values are used in executive dashboard KPIs. Both are accurate in their respective contexts. The key discipline is never mixing the two without explicit labelling.*

Exploratory Data Analysis

EDA was conducted systematically across four analytical areas: Portfolio Health and Risk Metrics (overall delinquency rates, exposure totals, segment distributions); Borrower Risk Analysis (segmentation by credit score band, DTI band, and risk segment); Portfolio Segmentation and Geographic Analysis (performance by property type, occupancy, loan purpose, and state); and Behavioural Risk Analysis (modification status, delinquency severity distribution, zero balance outcomes, and foreclosure activity).

For each dimension, both delinquency rates and exposure-weighted metrics were calculated. This dual approach was deliberate: delinquency rates identify where risk is most intense, but exposure-weighted metrics identify where risk is most financially material. These are often different segments — a finding that recurred consistently across the analysis and became a defining theme of the project.

Risk Segmentation Framework

The three-tier Risk Segment classification (Low Risk, Moderate Risk, High Risk) was central to the analytical approach. Rather than evaluating each risk factor independently, the risk-layering framework captured the compounding effect of multiple simultaneous risk conditions. This reflects the actual dynamics of mortgage default risk, which is rarely driven by a single factor and is typically the product of multiple interacting vulnerabilities.

Vintage analysis was conducted to evaluate delinquency performance across origination cohorts. This analysis is particularly important in the 2020 Q1–Q2 context, where older loans (2018–2019 vintages) had more seasoning and thus more opportunity to develop delinquency, while the newest 2020 Q2 vintage showed low delinquency rates that reflected insufficient seasoning rather than superior credit quality.

CREDIT RISK PERSPECTIVE *Vintage analysis is a core technique in mortgage portfolio surveillance. It allows credit risk teams to separate the effect of loan age (seasoning) from the effect of underlying credit quality when comparing cohort performance. A 2020 origination with 0.5% delinquency and a 2018 origination with 1.8% delinquency may actually have similar underlying credit quality — the difference is simply that the older loan has had more time to season into delinquency. This context is critical for accurate portfolio quality assessment.*

Target Variable Definition

The target variable for delinquency analysis was defined as any loan observation with a Current Loan Delinquency Status not equal to '00' (current). Serious delinquency was defined as status '03' or higher —

corresponding to 90 or more days past due — consistent with the Basel II/III definition of default and IFRS 9 Stage 3 classification criteria.

The 90+ DPD threshold was selected over the 30 DPD threshold for several reasons. First, it aligns with the regulatory default definition used in capital calculation. Second, 30 DPD delinquency has a high cure rate — a significant proportion of 30 DPD borrowers return to current status within one or two payment cycles, making the 30 DPD flag too noisy for default prediction. Third, 90+ DPD is the threshold that triggers loan loss reserve increases, formal workout programme consideration, and escalation to senior credit management — making it the most decision-relevant outcome variable.

10 DASHBOARD DESIGN PHILOSOPHY

Executive-First Design

The dashboard was designed for one primary audience: senior credit risk executives who need to understand portfolio risk quickly and act on it decisively. This meant prioritising immediate clarity over analytical comprehensiveness. Every design decision was evaluated against a single test: can a Chief Risk Officer extract the key insight from this visual in three seconds or less?

This executive-first orientation shaped every aspect of the dashboard — from the six-page structure and the KPI card ordering, to the chart type selection and the placement of the Executive Risk Commentary at the bottom of each page.

Six-Page Structure — Why This Number

Six pages were chosen because the project's findings naturally decomposed into six distinct management questions, and the dashboard's value depends entirely on each page adding new analytical value rather than repeating insights from previous pages.

The six pages follow a deliberate progressive narrative: what is happening (Page 1) → who is causing it (Page 2) → what structural factors amplify it (Page 3) → where it is concentrated and why (Page 4) → how distressed loans are behaving (Page 5) → what management should monitor going forward (Page 6). Each page answers one question and sets up the next.

Figure 3. Dashboard Storytelling Framework

A six-page narrative that takes executives from portfolio overview to granular risk insights and ongoing surveillance.



Figure 6. Dashboard Narrative Flow

Sequential analytical progression across six dashboard pages, demonstrating how each page answers one distinct executive business question and builds naturally on the findings of the previous page to create a coherent credit risk narrative.

Linear progression diagram showing the business question each page answers and the transition logic connecting consecutive pages. Illustrates how the dashboard builds from portfolio health through to operational surveillance.

Reporting Architecture — 1.92M vs 84M

The most consequential dashboard design decision was connecting the Power BI model to the latest-snapshot fact table (1.92M unique loans) rather than the full analytical panel (84.18M observations). This decision directly determined what every KPI card, every chart, and every filter on every page would show.

Connecting to the full panel would have produced misleading executive KPIs — a portfolio delinquency rate that double-counted cured loans, exposure figures inflated by cumulative monthly balances, and loan counts that reflected observations rather than unique borrowers. The snapshot model produces accurate current-state portfolio metrics that a CRO would recognise as meaningful.

PORTFOLIO MANAGEMENT NOTE *The decision to use a snapshot model rather than a panel for executive reporting is standard practice in institutional portfolio monitoring. Banks' risk dashboards show current outstanding positions — not the sum of all historical balances. Applying this principle to the analytical model was a deliberate choice to align the dashboard's outputs with the conventions executives actually use when evaluating portfolio health.*

Consistent KPI Strip

A design decision was made to maintain the same six KPI cards across all six dashboard pages. The alternative — page-specific KPI strips — was considered and rejected. Executive users navigating between analytical pages need a constant orientation anchor. Consistent KPIs across pages allow the executive to absorb changes in the analytical content while retaining the portfolio health baseline as a fixed reference point.

Colour Philosophy

The colour palette — navy blue, teal, gold, and white — was selected for institutional professional credibility rather than visual novelty. These colours are consistent with the Fannie Mae brand and the visual language of financial services reporting. Navy communicates authority and stability. Teal provides secondary visual hierarchy. Gold serves as a highlight and accent colour for thresholds and rule lines. White provides breathing space.

Visual Selection Principles

Chart type selection followed a strict business logic hierarchy: the chart type was selected based on what the executive needed to understand, not based on visual variety. Horizontal bar charts for ranked comparisons. Combo charts (bars + line) where two correlated dimensions needed simultaneous visibility. Shape maps for geographic pattern recognition. Conditional-formatted matrices for multi-dimensional surveillance scorecards. No chart was included because it looked interesting — each was included because it answered a specific executive question that no other visual could answer more efficiently.

Executive Risk Commentary

Every dashboard page includes an Executive Risk Commentary section at the bottom — a concise, business-focused narrative summarising the page's key findings in plain language. The commentary was designed to ensure that even an executive who does not fully engage with the individual visuals can extract the essential insight from one reading of the bottom panel. This mirrors the practice of real-world management reporting, where summaries and interpretive text accompany data visuals.

11 DASHBOARD WALKTHROUGH

This section documents each dashboard page in detail — its business purpose, the design rationale for each visual, the key insights delivered, and the analytical narrative that connects each page to the next.

Page 1 — Executive Portfolio Overview

Business Question: How healthy is the overall mortgage portfolio today?

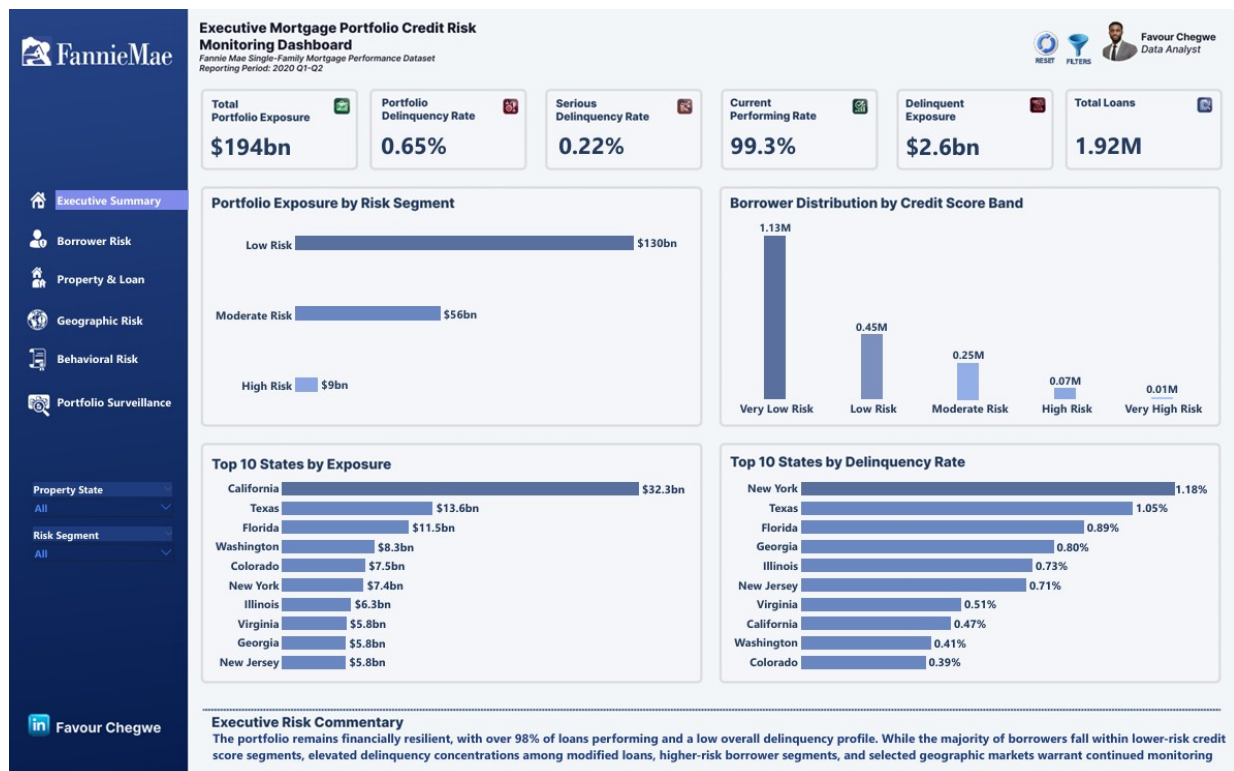


Figure 7. Page 1 — Executive Portfolio Overview

The dashboard's entry page presenting six portfolio-level KPIs, exposure distribution by risk segment, borrower credit score distribution, top 10 states by exposure, and top 10 states by delinquency rate — providing complete portfolio health orientation in a single view.

Screenshot of Page 1 showing the six KPI cards, Portfolio Exposure by Risk Segment, Borrower Distribution by Credit Score Band, Top 10 States by Exposure, and Top 10 States by Delinquency Rate.

Page 1 serves as the dashboard's entry point and executive orientation layer. Its purpose is not to provide deep analytical insight — that comes in Pages 2–5 — but to give an executive the complete portfolio health picture in a single page view. Every visual is chosen for speed of comprehension over analytical depth.

KPI Strip

Six cards present the portfolio's headline metrics: Total Portfolio Exposure (\$194bn), Portfolio Delinquency Rate (0.65%), Serious Delinquency Rate (0.22%), Current Performing Rate (99.3%), Delinquent Exposure (\$2.6bn), and Total Loans (1.92M). The ordering is intentional — exposure leads because it frames the financial scale, delinquency rates follow because they answer how stressed the portfolio is, delinquent exposure quantifies the financial risk in dollars, performing rate provides the reassurance, and total loans anchors the population count.

◆ Portfolio Exposure by Risk Segment

Business Question: How is portfolio financial exposure distributed across Low, Moderate, and High Risk borrowers?

Why This Visual: Horizontal bar chart — three segments with exposure values (\$130bn, \$56bn, \$9bn) allows immediate comparison of portfolio weight across risk categories at a glance.

Key Insight: The portfolio is structurally concentrated in the Low Risk segment (\$130bn), with Moderate Risk representing a meaningful secondary segment (\$56bn). High Risk exposure (\$9bn), while elevated in delinquency rate, represents a contained financial position.

Business Interpretation: Management can immediately see that the portfolio is not uniformly distributed — the High Risk tail is small in dollar terms, which shapes the proportionality of any intervention response. Large-scale policy changes are not warranted for a segment representing less than 5% of exposure.

◆ Borrower Distribution by Credit Score Band

Business Question: Where are the majority of borrowers positioned on the credit quality spectrum?

Why This Visual: Vertical bar chart — five bands from Very Low Risk to Very High Risk with loan counts. The declining volume from left to right communicates portfolio quality concentration at a glance without requiring any calculation.

Key Insight: 1.13 million loans (59% of portfolio) reside in the Very Low Risk band. Volume declines steeply with risk — only 10,000 loans in the Very High Risk band.

Business Interpretation: The portfolio's borrower quality distribution is a structural strength. Even if Very High Risk borrower delinquency rates are extreme, their small volume limits aggregate financial impact. This chart is the immediate reassurance visual that contextualises the risk findings that follow.

◆ Top 10 States by Portfolio Exposure

Business Question: Where is the portfolio's financial exposure geographically concentrated?

Why This Visual: Ranked horizontal bar chart — California leads at \$32.3bn. A choropleth map was evaluated and rejected for this use case: state size disparities would make small high-risk states nearly invisible while large, lower-risk states dominated the visual. The bar chart enables precise magnitude comparison across markets of very different size.

Key Insight: California alone accounts for approximately 16.6% of total exposure. Texas and Florida follow. The top three states collectively represent approximately 29% of total portfolio balance.

Business Interpretation: Geographic concentration of this magnitude creates systematic exposure to regional economic shocks. This chart plants the seed for the geographic risk analysis on Page 4 — the executive can immediately see where the financial weight sits before understanding the risk characteristics of those markets.

◆ Top 10 States by Delinquency Rate

Business Question: Which geographic markets exhibit the highest current delinquency rates?

Why This Visual: Ranked horizontal bar chart — New York leads at 1.18%. This chart was deliberately placed alongside the exposure chart rather than replacing it, because the two charts together tell a more important story than either alone.

Key Insight: New York (1.18%), Texas (1.05%), and Florida (0.89%) lead the delinquency ranking. California — despite leading in exposure — shows only 0.47% delinquency, well below the portfolio average.

Business Interpretation: The two geographic charts together reveal the key geographic insight: exposure concentration and credit risk concentration are different phenomena. California is a concentration risk, not a credit quality risk. This pairing on Page 1 sets up the diagnostic work of Page 4.

Page 1 Executive Commentary: The portfolio remains financially resilient, with over 98% of loans performing and a low overall delinquency profile. While the majority of borrowers fall within lower-risk credit score segments, elevated delinquency concentrations among modified loans, higher-risk borrower segments, and selected geographic markets warrant continued monitoring.

Transition to Page 2: Page 1 establishes that the portfolio is broadly healthy but that delinquency is not uniformly distributed. Page 2 answers the natural follow-up question: which specific borrower characteristics are driving the delinquency that does exist, and how much financial exposure is concentrated in those segments?

Page 2 — Borrower Risk Analysis

Business Question: Which borrowers are most likely to become delinquent, and where is delinquent exposure concentrated?

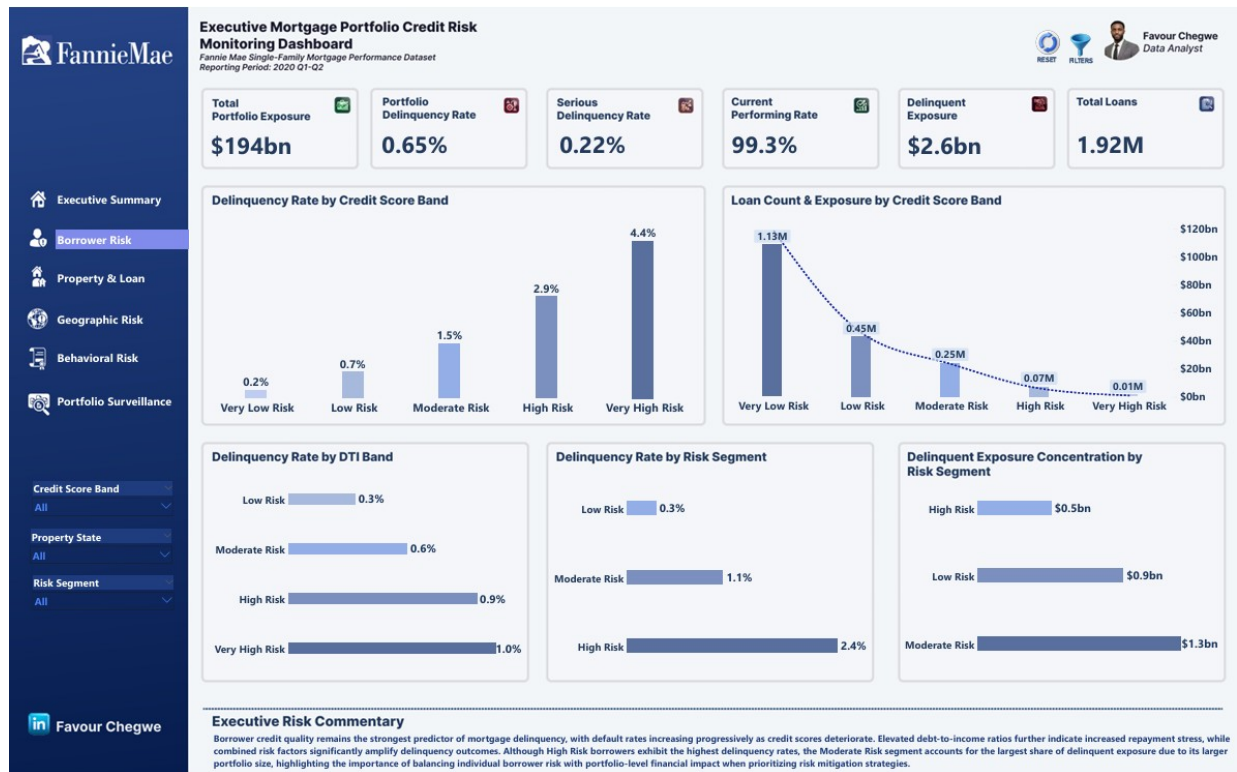


Figure 8. Page 2 — Borrower Risk Analysis

Five analytical visuals examining credit score delinquency gradient, loan count and exposure distribution, DTI risk relationships, risk segment delinquency rates, and delinquent exposure concentration — establishing that the Moderate Risk segment carries the largest delinquent exposure despite not having the highest delinquency rate.

Screenshot of Page 2 showing Delinquency Rate by Credit Score Band, Loan Count & Exposure by Credit Score Band, Delinquency Rate by DTI Band, Delinquency Rate by Risk Segment, and Delinquent Exposure Concentration by Risk Segment.

Page 2 moves from portfolio overview to borrower-level diagnosis. Every visual answers a specific question about one borrower characteristic and its relationship to delinquency. Critically, this page also addresses the financial exposure dimension — the Moderate Risk segment finding demonstrates that delinquency rates and delinquent exposure tell different stories about the same portfolio, a theme that became one of the project's defining insights.

◆ Delinquency Rate by Credit Score Band

Business Question: How dramatically does delinquency risk increase as borrower credit quality deteriorates?

Why This Visual: Vertical bar chart ordered from Very Low Risk (left) to Very High Risk (right) — the ascending bar heights create a visual representation of the risk gradient that communicates the relationship without requiring the reader to process the numbers first. The ordering was chosen deliberately so the escalation reads naturally left to right.

Key Insight: Delinquency rates: Very Low Risk 0.2%, Low Risk 0.7%, Moderate Risk 1.5%, High Risk 2.9%, Very High Risk 4.4%. A 22-fold difference between the extremes.

Business Interpretation: Credit score is the single most powerful underwriting variable in this portfolio. This chart is the empirical foundation for credit score-based risk segmentation in underwriting policy.

◆ Loan Count and Exposure by Credit Score Band

Business Question: Where is the portfolio's financial weight positioned relative to the delinquency risk gradient?

Why This Visual: Combo chart — bars for loan count, dotted line for exposure. Both series decline from left (Very Low Risk) to right (Very High Risk), showing that most loans and most money are in the better-performing segments. An alternative of two separate charts was rejected — showing both dimensions on one chart is more powerful for the management communication objective.

Key Insight: 1.13M loans and the majority of exposure sit in the Very Low Risk band. The High Risk and Very High Risk bands, despite elevated delinquency, hold only 80,000 loans between them.

Business Interpretation: This chart answers the implicit follow-up to every delinquency rate chart: how many borrowers are actually in that high-risk segment? The answer reassures management that the high-risk tail, while severe in rate, is small in financial scale.

◆ Delinquency Rate by DTI Band

Business Question: Does payment capacity — measured by debt-to-income ratio at origination — independently predict delinquency?

Why This Visual: Horizontal bar chart ordered from Low Risk DTI to Very High Risk DTI — a clean directional comparison that confirms the relationship without overstating its magnitude.

Key Insight: DTI delinquency rates: Low Risk 0.3%, Moderate Risk 0.6%, High Risk 0.9%, Very High Risk 1.0%. Consistent directional relationship, moderate differentiation.

Business Interpretation: DTI is a secondary risk driver. It confirms the directional relationship between payment capacity and delinquency, but the differentiation is less dramatic than credit score. Its strongest contribution to risk management is through the risk-layering framework rather than as a standalone underwriting threshold.

◆ Delinquency Rate by Risk Segment

Business Question: What happens to delinquency risk when multiple risk factors are present simultaneously?

Why This Visual: Horizontal bar chart with three segments — Low (0.3%), Moderate (1.1%), High (2.4%). The gap between Low and High quantifies the compounding effect of risk-layering.

Key Insight: High Risk borrowers are delinquent at a rate eight times that of Low Risk borrowers. The jump from Moderate to High is more pronounced than from Low to Moderate — indicating non-linear compounding.

Business Interpretation: Single-variable underwriting misses this compounding effect entirely. Risk-layering analysis provides the empirical basis for requiring compensating factors or reduced LTV when a borrower exhibits multiple concurrent risk conditions.

◆ Delinquent Exposure Concentration by Risk Segment

Business Question: Which risk segment accounts for the largest share of delinquent financial exposure?

Why This Visual: Horizontal bar chart — Low Risk \$0.9bn, Moderate Risk \$1.3bn, High Risk \$0.5bn. A donut chart was evaluated and rejected — donut charts struggle to communicate absolute dollar values accurately, and the absolute amounts are more important than percentage shares for management decision-making.

Key Insight: Moderate Risk carries the highest delinquent exposure (\$1.3bn) despite not having the highest delinquency rate — because it contains far more loans than the High Risk segment.

Business Interpretation: This is the page's most analytically sophisticated finding: the segment with the worst delinquency rate is not the segment with the most money at risk. This recurs across multiple dimensions of the project and has significant implications for how surveillance resources should be allocated.

Page 2 Executive Commentary: Borrower credit quality remains the strongest predictor of mortgage delinquency. The Moderate Risk segment accounts for the largest share of delinquent exposure due to its significant loan volume, demonstrating that effective risk management requires weighting both delinquency intensity and outstanding balance rather than rate alone.

Transition to Page 3: Page 2 established who is risky — the borrower-level risk drivers. Page 3 examines what they borrowed — the loan structures and property characteristics that create structural risk independent of borrower quality.

Page 3 — Property and Loan Characteristics

Business Question: Which loan and property structures carry elevated delinquency risk, and do they represent a material portfolio concentration?

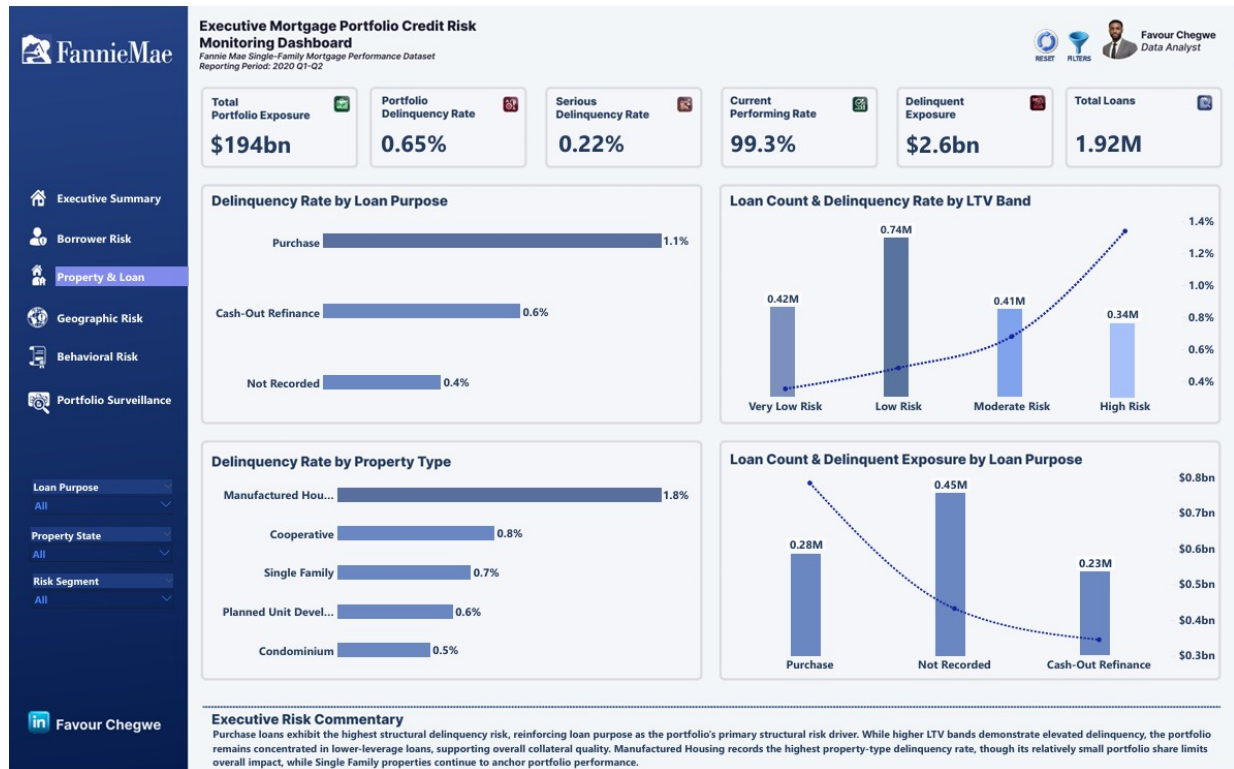


Figure 9. Page 3 — Property and Loan Characteristics

Four visuals covering loan purpose delinquency differentials, LTV band risk and concentration, property type performance, and loan purpose exposure — confirming purchase loans as the primary structural risk driver and the portfolio's concentration in lower-LTV positions as a collateral quality strength.

Screenshot of Page 3 showing Delinquency Rate by Loan Purpose, Loan Count & Delinquency Rate by LTV Band, Delinquency Rate by Property Type, and Loan Count & Delinquent Exposure by Loan Purpose.

Page 3 shifts the analytical lens from borrower characteristics to structural and collateral characteristics. The key distinction from Page 2 is that structural risk is largely fixed at origination — a 30-year fixed rate mortgage originated at 90% LTV will carry that structural risk profile for its entire life, regardless of how the borrower's circumstances change. Understanding structural risk is therefore critical for prospective portfolio composition decisions, not only retrospective monitoring.

◆ Delinquency Rate by Loan Purpose

Business Question: Do loans originated for different purposes exhibit meaningfully different delinquency outcomes?

Why This Visual: Horizontal bar chart — Purchase (1.1%), Cash-Out Refinance (0.6%), Not Recorded (0.4%). Ordered by delinquency rate descending so the highest-risk purpose is immediately visible. The bar format was chosen because loan purpose is a categorical variable with a small number of meaningful categories — it communicates comparison more efficiently than any alternative visual.

Key Insight: Purchase loans are approximately double the delinquency rate of Cash-Out Refinance loans. For the 2020 Q1-Q2 period, purchase loans originated closest to the COVID-19 economic shock during the most financially vulnerable early months of the loan lifecycle.

Business Interpretation: Loan purpose warrants inclusion as a distinct monitoring dimension in portfolio reporting. Purchase loan cohorts from 2020 require vintage-adjusted tracking to distinguish COVID-timing effects from structural loan-purpose risk.

◆ Loan Count and Delinquency Rate by LTV Band

Business Question: How does borrower leverage position affect delinquency risk, and where is the portfolio concentrated by LTV?

Why This Visual: Combo chart — bars for loan count per LTV band, dotted line for delinquency rate. The same two-dimension logic applied in Page 2's credit score combo chart applies here: showing both the risk gradient and the portfolio concentration simultaneously delivers twice the management insight in a single visual.

Key Insight: Delinquency rises from 0.4% in Very Low Risk LTV to 1.4% in High Risk LTV. Portfolio concentration in the Low LTV band (0.74M loans) provides a structural buffer. The High Risk LTV band (0.34M loans) shows the steepest delinquency rate.

Business Interpretation: The portfolio's concentration in lower-LTV bands is a collateral quality strength. Low LTV means more equity — more buffer between the loan balance and the property value — which directly reduces both the probability of strategic default and the loss severity in a default scenario.

◆ Delinquency Rate by Property Type

Business Question: Does the type of property collateral affect delinquency outcomes?

Why This Visual: Horizontal bar chart ordered by delinquency rate — Manufactured Housing (1.8%), Cooperative (0.8%), Single Family (0.7%), Planned Unit Development (0.6%), Condominium (0.5%). The ranking order was chosen deliberately to make the Manufactured Housing finding immediately visible.

Key Insight: Manufactured Housing exhibits a delinquency rate 3.6 times that of Condominiums. Single Family properties — the dominant collateral type — perform at the portfolio average.

Business Interpretation: Property type differentials reflect differences in collateral liquidity, resale value stability, and borrower financial profile. Manufactured Housing warrants additional underwriting scrutiny. The analysis was originally considered for replacement with a matrix format, but the post-snapshot rebuild showed sufficient variation to justify a dedicated visual.

◆ Loan Count and Delinquent Exposure by Loan Purpose

Business Question: Does the purchase loan delinquency finding translate into material delinquent exposure in dollar terms?

Why This Visual: Combo chart — bars for loan count, line for delinquent exposure by loan purpose. This chart completes the loan purpose analysis by adding financial weight to the rate finding from Chart 1. The decision to show loan count as bars and delinquent exposure as a line was intentional — it allows the executive to see both portfolio volume and financial risk simultaneously without requiring two separate visuals.

Key Insight: Purchase loans carry the highest delinquent exposure (\$0.7bn) consistent with their elevated delinquency rate. Cash-Out Refinance follows with \$0.3bn.

Business Interpretation: The structural risk story for loan purpose is now complete: purchase loans are not only more frequently delinquent — they also account for the largest concentration of delinquent dollars, confirming them as the primary structural risk driver and warranting separate monitoring.

Page 3 Executive Commentary: Purchase loans exhibit the highest structural delinquency risk. LTV analysis confirms that the portfolio's concentration in lower-leverage positions provides collateral quality support. Manufactured Housing records the highest property-type delinquency rate, though its limited portfolio share constrains aggregate financial impact.

Transition to Page 4: Pages 2 and 3 answered who and what. Page 4 asks where — specifically, whether the geographic markets with the most exposure also carry the most credit risk, and whether that risk reflects location itself or the underlying borrower composition within each market.

Page 4 — Geographic Risk Analysis

Business Question: Why are certain geographic markets riskier than others?

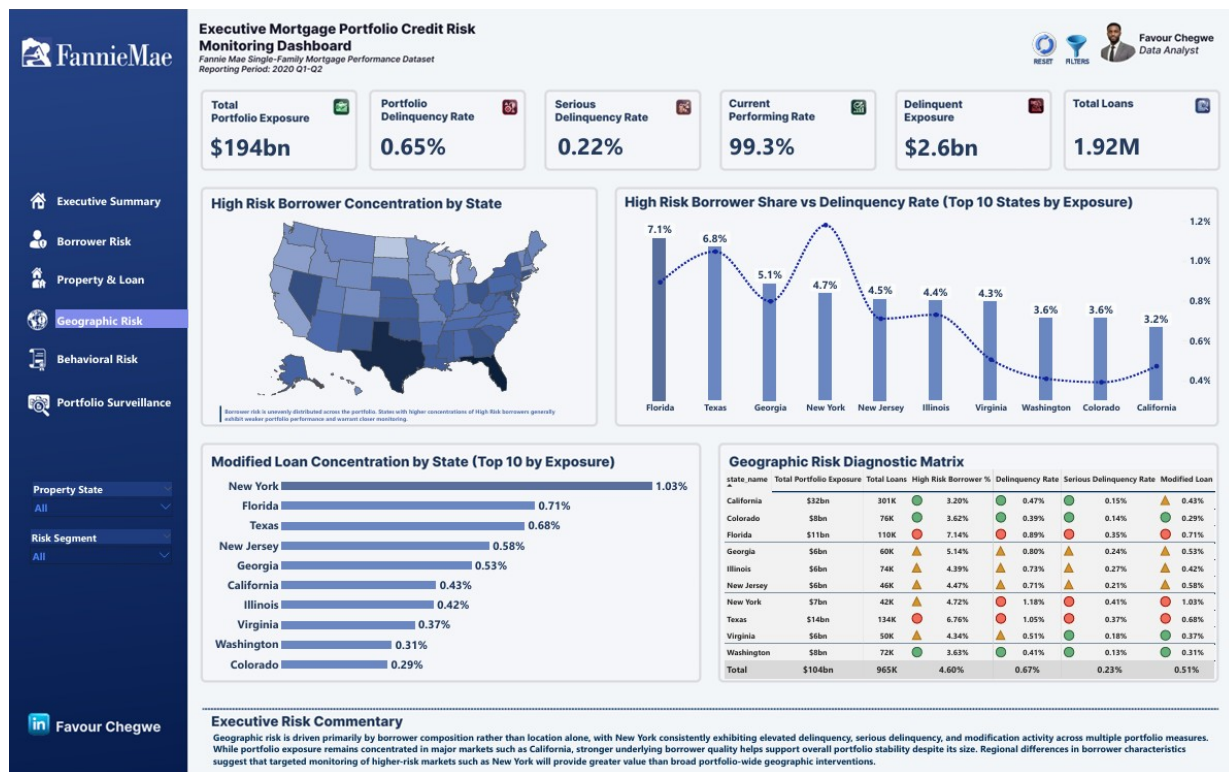


Figure 10. Page 4 — Geographic Risk Analysis

Four diagnostic visuals — choropleth map, High Risk borrower share vs delinquency rate combo chart, modified loan concentration ranking, and conditional-formatted diagnostic matrix — confirming that geographic risk is driven by borrower composition rather than location alone, with New York elevated across all five monitored risk dimensions.

Screenshot of Page 4 showing the High Risk Borrower Concentration Shape Map, High Risk Borrower Share vs Delinquency Rate combo chart, Modified Loan Concentration horizontal bar, and the Geographic Risk Diagnostic Matrix with conditional formatting.

Page 4 is the dashboard's most analytically sophisticated page. Rather than simply ranking states by delinquency rate — which Page 1 already partially achieves — Page 4 diagnoses the drivers of geographic risk. The central hypothesis being tested is: do states exhibit elevated delinquency because of location-specific economic factors, or because they contain higher concentrations of risky borrower profiles? The answer shapes the policy response entirely — and the data provides a definitive answer.

KEY INSIGHT The diagnostic reframe of Page 4 — from 'which states are risky' to 'why are certain states riskier' — was the single most important design evolution in the dashboard development process. A ranking page answers a question executives can partially answer themselves. A diagnostic page answers a question they cannot answer without the analysis.

◆ High Risk Borrower Concentration by State (Shape Map)

Business Question: Is High Risk borrower concentration geographically distributed in a pattern that correlates with state-level delinquency?

Why This Visual: Choropleth map coloured by High Risk borrower share — darker shading indicates higher concentration. The map was retained despite the earlier consideration of overall delinquency rate as the colouring variable: using High Risk borrower share directly tests the compositional hypothesis. The map serves as the entry point that establishes the geographic pattern before the diagnostic charts below explain it.

Key Insight: High Risk borrower concentration varies across states — Florida (7.14%) and Texas (6.76%) show higher concentrations than California (3.20%) and Washington (3.63%). The pattern partially mirrors state-level delinquency rates.

Business Interpretation: The map directs executive attention toward states with high High Risk borrower concentrations, setting up the analytical question that the next visual answers: does higher concentration actually translate into higher delinquency?

◆ High Risk Borrower Share vs Delinquency Rate — Top 10 States by Exposure

Business Question: Do states with higher concentrations of High Risk borrowers also exhibit higher delinquency rates?

Why This Visual: Combo chart — bars for High Risk borrower share per state, dotted line for state delinquency rate. States are ordered by exposure (California to Washington) rather than by risk, so the visual directly shows the risk profile of the portfolio's most financially significant markets.

Key Insight: Florida (7.14% High Risk share, 0.89% DQ), Texas (6.76%, 1.05% DQ), and New York (4.72%, 1.18% DQ) show above-average readings on both dimensions. California (3.20%, 0.47% DQ) shows low High Risk share and below-average delinquency despite leading in exposure.

Business Interpretation: The visual confirms the diagnostic hypothesis: geographic risk tracks borrower quality composition. California's large exposure is not a credit quality risk because its borrower quality is comparatively strong. New York's elevated delinquency is explained by its elevated High Risk borrower concentration.

◆ Modified Loan Concentration by State — Top 10 by Exposure

Business Question: Are high-delinquency states also states where loan modification activity is elevated — providing a second independent confirmation of the risk profile?

Why This Visual: Ranked horizontal bar chart — New York leads at 1.03% modification rate, more than double the next highest state. Ordering by modification rate rather than exposure allows the outlier nature of New York's reading to be immediately visible.

Key Insight: New York's modification concentration (1.03%) is dramatically higher than all other major markets. Florida (0.71%) and Texas (0.68%) follow. California (0.43%) and Washington (0.31%) show below-average modification rates consistent with their strong credit quality.

Business Interpretation: New York's co-occurrence of the highest delinquency rate AND the highest modification rate across two independent metrics is a critical finding. It confirms that New York's risk profile is systemic rather than driven by a single factor.

◆ Geographic Risk Diagnostic Matrix

Business Question: What is the complete multi-dimensional risk profile of each major geographic market simultaneously?

Why This Visual: Conditional-formatted matrix — 10 states × 6 metrics (Exposure, Loans, High Risk %, DQ Rate, Serious DQ Rate, Modified Loan %) with RAG colour coding applied per column based on portfolio-relative thresholds. This is the only visual format that communicates the complete risk profile of every material market in a single view. No chart equivalent exists that can show six dimensions across ten entities with comparable clarity.

Key Insight: New York is the only state to show elevated readings across all five risk metrics simultaneously. California shows green or amber across almost all metrics despite its dominant exposure. Florida and Texas show multiple red/amber readings.

Business Interpretation: The matrix is the executive synthesis of the entire geographic analysis: one table that communicates the complete risk profile of every material market. It proves that exposure concentration and credit risk concentration are different phenomena — the most important geographic insight in the project.

Page 4 Executive Commentary: Geographic risk is driven by borrower composition rather than location alone. New York consistently exhibits elevated delinquency, serious delinquency, and modification activity across multiple independent measures. California's strong borrower quality supports overall portfolio stability despite its scale.

Transition to Page 5: Page 4 confirmed that modified loans represent a disproportionate risk driver across geographic markets — particularly New York. Page 5 isolates modification status as a behavioural risk dimension in its own right, testing whether the modification effect operates independently of the borrower quality classification already established in Page 2.

Page 5 — Behavioural Risk Analysis

Business Question: How does modification status influence delinquency severity, and does this effect persist independently of borrower risk classification?

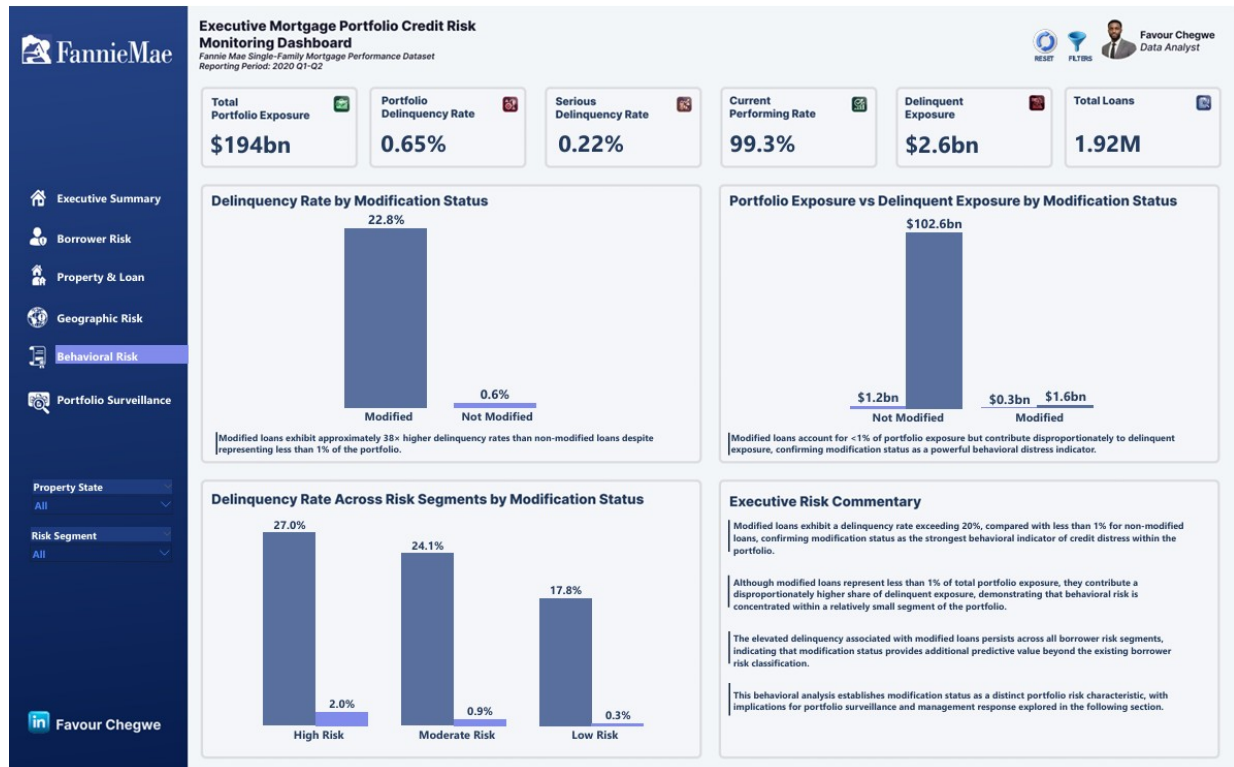


Figure 11. Page 5 — Behavioural Risk Analysis

Three visuals establishing modification status as the portfolio's single strongest independent behavioural risk predictor — with a 22.8% vs 0.6% delinquency differential that persists across all borrower risk segments, confirming that modification status carries risk information independent of existing risk classification.

Screenshot of Page 5 showing Delinquency Rate by Modification Status, Portfolio Exposure vs Delinquent Exposure by Modification Status, and Delinquency Rate Across Risk Segments by Modification Status.

Page 5 is the dashboard's analytical centrepiece — the page that contains the project's single most striking finding. The entire page is built around one diagnostic question: is modification status an independent risk predictor, or does it simply proxy for the pre-existing risk characteristics already captured in the borrower segmentation?

This distinction matters enormously for surveillance design. If modification merely proxies for high-risk borrowers, then monitoring the High Risk segment is sufficient. If modification provides independent predictive value beyond risk segment classification, then a dedicated modification monitoring programme is necessary — one that tracks modified loans separately regardless of their original risk segment. The data provides an unambiguous answer.

◆ Delinquency Rate by Modification Status

Business Question: How dramatically does modification status affect a loan's current likelihood of delinquency?

Why This Visual: Two-bar comparison chart — Modified (22.8%) vs Not Modified (0.6%). The format prioritises magnitude communication over fine-grained comparison. Two large bars with clearly labelled values communicate the scale of the differential more powerfully than a more complex visual would.

Key Insight: A 38-fold delinquency rate differential between modified and non-modified loans. More than one in five modified loans is currently delinquent.

Business Interpretation: This is the project's headline behavioural finding. Modified loans have already demonstrated one episode of financial distress severe enough to require formal loan restructuring — and the data shows that this distress is not resolved by modification alone. Modified loans remain the highest-risk behavioural population in the portfolio.

◆ Portfolio Exposure vs Delinquent Exposure by Modification Status

Business Question: Does the modification delinquency finding translate into a disproportionate concentration of delinquent financial exposure?

Why This Visual: Four-bar comparison — Non-Modified total exposure (\$102.6bn), Non-Modified delinquent exposure (\$1.2bn), Modified total exposure (\$0.3bn), Modified delinquent exposure (\$1.6bn). Actual dollar values were used rather than percentage shares — the absolute amounts are more management-relevant than percentage proportions for exposure assessment.

Key Insight: Modified loans hold \$0.3bn in total exposure but \$1.6bn in delinquent exposure — more than five times their total balance. This means the modified loan segment's delinquent exposure exceeds its total portfolio exposure by a factor of five.

Business Interpretation: This is the financial proof of the concentration argument. Modification status identifies a tiny segment that generates delinquent exposure wildly disproportionate to its portfolio weight. This finding directly supports Recommendation 1: dedicated modification monitoring delivers more risk management value per dollar of resource than any other monitoring investment.

◆ Delinquency Rate Across Risk Segments by Modification Status

Business Question: Does modification status add delinquency risk above and beyond what is captured by the existing borrower risk segment classification?

Why This Visual: Clustered bar chart — for each of three Risk Segments (Low, Moderate, High), two bars showing modified versus non-modified delinquency rates side by side. This was the page's most analytically critical visual choice: showing the modification effect within each risk segment is the only way to test whether modification operates independently of pre-existing risk classification.

Key Insight: High Risk Modified: 27.0% vs High Risk Not Modified: 2.0%. Moderate Risk Modified: 24.1% vs Not Modified: 0.9%. Low Risk Modified: 17.8% vs Not Modified: 0.3%. The modification premium exists in every segment.

Business Interpretation: This chart proves independence conclusively: a Low Risk modified borrower (17.8% DQ) is approximately 59 times more likely to be delinquent than a Low Risk non-modified borrower (0.3% DQ). Modification status is not a proxy for risk segment — it is an independent risk signal that must be tracked separately.

Page 5 Executive Commentary: Modified loans exhibit a delinquency rate exceeding 20%, compared with less than 1% for non-modified loans. This risk persists across all borrower risk segments, indicating that modification status provides additional predictive value beyond existing borrower risk classification. Implications for portfolio surveillance and management response are addressed in the following section.

Transition to Page 6: Pages 1–5 established what the portfolio looks like, where the risk lives, who is carrying it, what structural factors amplify it, and how distressed loans are behaving. Page 6 translates all of those findings into an operational monitoring framework with defined thresholds, escalation status, and management priorities.

Page 6 — Portfolio Surveillance

Business Question: Which portfolio segments require continuous management attention, and which are currently in breach of risk thresholds?

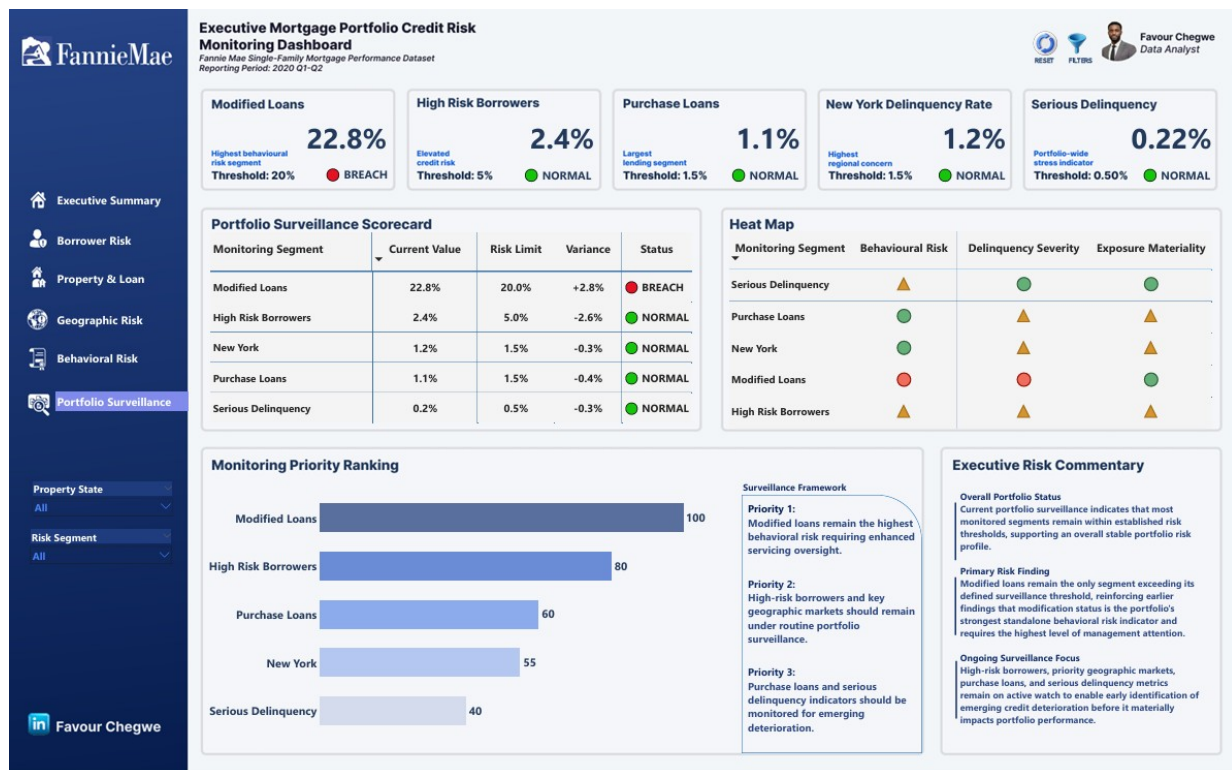


Figure 12. Page 6 — Portfolio Surveillance

Operational monitoring page featuring five Active Monitoring Priority Cards, Portfolio Surveillance Scorecard, Risk Heat Map, Monitoring Priority Ranking, and Surveillance Framework — confirming Modified Loans as the only segment in breach of its defined threshold at 22.8% against a 20.0% risk limit.

Screenshot of Page 6 showing the five Active Monitoring Priority Cards with RAG status, Portfolio Surveillance Scorecard, Risk Heat Map, Monitoring Priority Ranking, and Surveillance Framework text panel.

Page 6 is structurally different from Pages 1–5. Where previous pages were analytical — presenting findings, establishing patterns, diagnosing causes — Page 6 is operational. Its purpose is to translate the analytical work of the entire dashboard into a management framework that can be implemented, maintained, and reported on regularly.

The deliberate replacement of the standard KPI strip with five Active Monitoring Priority Cards signals this mode change. The executive has moved from analysis to action. The monitoring cards, scorecard, heatmap, and priority ranking are not new analytical findings — they are the institutionalisation of the findings from Pages 1–5 into an operational monitoring framework.

◆ Active Monitoring Priority Cards

Business Question: Which specific segments are currently in breach of defined risk thresholds, and which are within normal limits?

Why This Visual: Five large KPI-style cards replace the standard portfolio KPI strip used on Pages 1–5. Each card shows: segment name, current value, risk threshold, and RAG status. This format signals mode change from analytical reporting to operational monitoring — a design decision that mirrors how risk committee reports are typically structured in banking institutions.

Key Insight: Modified Loans: 22.8% against 20.0% threshold — BREACH (+2.8%). High Risk Borrowers: 2.4% against 5.0% — NORMAL. New York: 1.2% against 1.5% — NORMAL. Purchase Loans: 1.1% against 1.5% — NORMAL. Serious Delinquency: 0.2% against 0.5% — NORMAL.

Business Interpretation: At a single glance, the CRO knows: one segment is in breach, four are within limits, and the overall portfolio surveillance status is stable with one active escalation required. No analysis needed — the operational status is immediately apparent.

◆ Portfolio Surveillance Scorecard

Business Question: What is the precise current value, risk limit, variance, and status of every monitored segment?

Why This Visual: Conditional-formatted matrix with four columns: Current Value, Risk Limit, Variance, and Status. The matrix format was chosen specifically because it mirrors standard risk committee reporting formats in financial institutions. A chart cannot communicate threshold comparison, variance magnitude, and status simultaneously — only a formatted table can.

Key Insight: Modified Loans variance of +2.8% is the only positive variance in the scorecard — the only segment where current performance exceeds its threshold. All other segments show negative variance, confirming compliance with monitoring limits.

Business Interpretation: The scorecard provides the precision layer behind the monitoring cards. It translates the broad surveillance picture into the specific numbers a risk committee needs to record, track, and report against.

◆ Risk Heat Map

Business Question: Across three risk dimensions — behavioural risk, delinquency severity, and exposure materiality — which monitoring segments show multi-dimensional elevation?

Why This Visual: Matrix of monitoring segments × risk dimensions with RAG indicators per cell. The heat map was included because it communicates multi-dimensional risk profiles across all five monitored segments simultaneously — something a scorecard cannot do efficiently with additional columns.

Key Insight: Modified Loans shows red across Behavioural Risk and Delinquency Severity. New York and Purchase Loans show amber in Delinquency Severity and Exposure Materiality. High Risk Borrowers shows amber across all three dimensions.

Business Interpretation: The heat map reveals nuance that the scorecard cannot: while Modified Loans is the only threshold breach, New York and Purchase Loans carry multi-dimensional amber signals suggesting elevated future risk that warrants active monitoring before a threshold breach occurs.

◆ Monitoring Priority Ranking

Business Question: How should management rank the five monitored segments for resource allocation purposes?

Why This Visual: Horizontal bar chart with indexed priority scores: Modified Loans (100), High Risk Borrowers (80), Purchase Loans (60), New York (55), Serious Delinquency (40). An indexed format was chosen over raw delinquency rates to communicate relative priority rather than absolute risk levels — the distinction between Priority 1 and Priority 5 is the management signal, not the specific delinquency rate.

Key Insight: Modified Loans receives the maximum priority score. High Risk Borrowers ranks second due to their 2.4% delinquency rate, which represents the second-highest absolute risk level even while within the 5.0% threshold.

Business Interpretation: The priority ranking provides management with an explicit resource allocation framework. If monitoring budgets are constrained, this chart answers the question of where to concentrate oversight effort for the highest risk management return.

Page 6 Executive Commentary: Modified loans remain the only segment exceeding its defined surveillance threshold, reinforcing modification status as the portfolio's strongest standalone behavioural risk indicator. All other monitored segments remain within established limits, supporting an overall stable portfolio risk profile.

EXECUTIVE TAKEAWAY *Page 6 is the operational conclusion of a six-page analytical journey. Every monitoring threshold, every priority score, and every RAG indicator on this page is grounded in the findings of Pages 1–5. The surveillance framework has analytical authority because the analysis behind it is rigorous, documented, and traceable.*

12 OVERALL BUSINESS FINDINGS

Consolidating the findings across all six dashboard pages, the mortgage portfolio analysis produced eight principal business conclusions:

Finding 1 — The Portfolio Is Fundamentally Healthy

Observation: With 99.3% of loans performing as contractually expected and a delinquency rate of 0.65%, the portfolio demonstrates sound overall credit quality.

Interpretation: The portfolio's borrower quality composition — nearly 59% in the Very Low Risk credit score band — provides structural protection against broad-based deterioration.

Impact: No systemic credit action is warranted. The portfolio's risk profile supports a maintain-and-monitor posture.

Executive Decision: Continue standard portfolio monitoring with targeted escalation for identified high-risk segments.

Finding 2 — Credit Score Is the Dominant Individual Risk Driver

Observation: A 22-fold difference in delinquency rates exists between the Very Low Risk and Very High Risk credit score segments.

Interpretation: Credit score is not merely correlated with delinquency — it is the strongest single predictor in the portfolio, validated consistently across all analytical dimensions.

Impact: Credit score cutoffs and segment-specific monitoring thresholds are the highest-value underwriting tools available.

Executive Decision: Maintain credit score as the primary origination screening variable and segment-level monitoring anchor.

Finding 3 — Risk Layering Amplifies Delinquency Non-Linearly

Observation: High Risk layered borrowers exhibit delinquency at 2.4% versus 0.3% for Low Risk borrowers — an eight-fold difference.

Interpretation: The combination of multiple risk conditions produces non-linear amplification that single-variable underwriting frameworks fail to capture.

Impact: Single-variable threshold-based underwriting materially underestimates the risk of multi-factor borrowers.

Executive Decision: Implement risk-layering policy requiring additional compensating factors when two or more risk conditions are simultaneously present.

Finding 4 — Modification Status Is the Strongest Behavioural Predictor

Observation: Modified loans exhibit 22.8% delinquency versus 0.6% for non-modified — a 38-fold differential that persists across all risk segments.

Interpretation: Modification status is an independent risk signal, not merely a proxy for pre-existing borrower weakness. A Low Risk modified borrower carries nearly 60 times the delinquency risk of a Low Risk non-modified borrower.

Impact: Modification status must be tracked as a primary portfolio risk variable in its own dedicated monitoring framework.

Executive Decision: Establish a dedicated monthly modified loan monitoring programme with escalation to workout teams at 60+ DPD post-modification.

Finding 5 — Purchase Loans Carry Structural Risk

Observation: Purchase loans record delinquency rates approximately double those of refinance loans (1.1% vs 0.6%).

Interpretation: This differential is consistent with vintage and seasoning effects specific to the 2020 COVID period, but requires longitudinal validation to distinguish timing from structural risk.

Impact: Purchase loan performance must be tracked separately from refinance loans, disaggregated by origination vintage.

Executive Decision: Implement vintage-adjusted purchase loan monitoring. Initiate underwriting review if elevated delinquency persists beyond 2020 cohorts.

Finding 6 — Geographic Risk Reflects Borrower Composition

Observation: New York (1.18% DQ, 1.03% modification) leads across multiple independent risk dimensions. California (0.47% DQ) performs below the portfolio average despite 16.6% exposure concentration.

Interpretation: State-level delinquency variation is explained by differences in underlying borrower quality rather than location-specific economic conditions.

Impact: Geographic risk management requires different responses for different markets: credit quality monitoring for high-delinquency states; scenario stress testing for high-exposure states.

Executive Decision: Increase monitoring frequency for New York; develop HPI stress scenarios for California.

Finding 7 — Exposure and Delinquency Rate Tell Different Stories

Observation: The Moderate Risk segment carries \$1.3bn in delinquent exposure versus \$0.5bn for the High Risk segment, despite lower delinquency rates.

Interpretation: Volume differences between segments dominate individual rate differences at portfolio scale. Delinquency-rate-only monitoring systematically misallocates oversight resources.

Impact: Exposure-weighted monitoring metrics must accompany delinquency rates in all portfolio reporting.

Executive Decision: Mandate dual reporting of delinquency rate and delinquent exposure for all monitored segments.

Finding 8 — The Portfolio Is in the COVID-19 Policy Shadow

Observation: Foreclosure activity is near-zero; forbearance classified stressed borrowers as current during the reporting period.

Interpretation: Reported delinquency metrics represent a floor of actual borrower stress. True portfolio performance will only be fully visible as forbearance programmes expire.

Impact: All current-period performance metrics should be treated as understated relative to underlying credit stress.

Executive Decision: Build forbearance-adjustment methodology for subsequent reporting periods as assistance programmes unwind.

13 BUSINESS IMPACT

Decision Support Value

This analysis translates raw mortgage performance data into five distinct categories of management decision support:

Underwriting Policy

The risk driver analysis — particularly the credit score gradient and risk-layering findings — provides quantitative evidence that can be used to calibrate underwriting cutoffs and compensating factor requirements. The finding that Very High Risk borrowers exhibit delinquency 22 times higher than Very Low Risk borrowers supports credit score minimums. The risk-layering analysis supports policies requiring stronger compensating factors when multiple risk conditions are simultaneously present.

Collections Resource Allocation

By identifying that modification status is the strongest behavioural predictor of delinquency — independent of original risk segment — the analysis enables collections teams to prioritise outreach more efficiently. Modified loans in any risk segment are more than 20 times more likely to be delinquent than non-modified loans. Collections outreach targeted at modified loans will produce more recoveries per dollar of resource expenditure than broad portfolio-wide campaigns.

Portfolio Monitoring

The six-segment surveillance framework with defined thresholds provides portfolio managers with an operational monitoring structure that can be maintained and reported on a periodic basis. Rather than reanalysing the entire portfolio each quarter from scratch, managers can track threshold compliance across the five priority segments and escalate only when a segment crosses its limit.

Geographic Stress Testing

The geographic diagnostic matrix provides the starting point for regional stress testing exercises. The identification of California as an exposure concentration risk — and the finding that its risk is driven by scale rather than credit quality — supports scenario analyses focused on California house price movements and their impact on portfolio LTV distributions and expected delinquency uplift.

Executive Communication

The six-page dashboard provides a structured, consistent framework for regular executive portfolio updates. Rather than requiring ad-hoc analysis for each committee meeting, the dashboard enables a standardised reporting cycle in which the same analytical framework is refreshed with updated data, allowing executives to track trends over time rather than evaluating each quarter's findings in isolation.

14 STRATEGIC RECOMMENDATIONS

The five recommendations below are derived directly from the project's analytical findings. Each is framed around the business reasoning that supports it, not merely the statistical finding that prompted it.

Recommendation 1 — Establish a Dedicated Modified Loan Monitoring Programme

Business reasoning: The 22.8% delinquency rate in the modified loan segment is the project's most actionable finding. Modified loans have already experienced one episode of financial distress severe enough to require formal restructuring. The data shows that even after modification, their delinquency risk remains extreme — and that this risk exists independently of their original borrower quality tier. This is not a segment that benefits from standard portfolio monitoring; it requires dedicated case-level oversight.

Implementation: Monthly servicer-level review of all modified loans. Escalation to workout teams when a modified loan reaches 60+ DPD post-modification. Quarterly reporting of re-delinquency rates by modification type to the Credit Risk Committee.

CREDIT RISK PERSPECTIVE *Modified loan re-default rates are a standard performance metric in bank loan modification programme evaluation. Tracking re-default rates by modification type (rate reduction, term extension, principal deferral) allows management to identify which modification structures are most effective at producing durable cures — critical information for optimising loss mitigation programmes.*

Recommendation 2 — Strengthen Risk-Layered Borrower Oversight

Business reasoning: The risk-layering analysis demonstrates that the combination of multiple risk factors creates exponentially worse outcomes than any single factor in isolation. Standard single-variable underwriting thresholds fail to capture this compounding dynamic.

Implementation: Introduce risk-layering requirements into the underwriting policy framework. Borrowers exhibiting two or more concurrent risk conditions should face additional compensating factor requirements or reduced LTV limits. Portfolio monitoring should track the High Risk and Moderate Risk segments by both delinquency rate and exposure share monthly.

Recommendation 3 — Implement Vintage-Adjusted Purchase Loan Monitoring

Business reasoning: The 1.1% delinquency rate on purchase loans — double the refinance rate — is a significant finding that requires careful interpretation. The 2020 Q1-Q2 period placed purchase loan originations directly in the path of the COVID-19 economic shock during the most financially vulnerable early months of the mortgage lifecycle. Until vintage-adjusted analysis across subsequent reporting periods is available, treating this as a known risk that requires tracking is the prudent approach.

Implementation: Separate purchase loan tracking from refinance loan tracking in all portfolio reports. Disaggregate by origination quarter. If vintage-adjusted purchase loan delinquency remains materially elevated beyond 2020 cohorts, initiate a formal underwriting review.

Recommendation 4 — Geographic Stress Monitoring for Priority Markets

Business reasoning: The geographic diagnostic analysis reveals two distinct types of geographic risk that require different management responses. California's risk is an exposure concentration risk. New York's risk is a credit quality risk — its borrowers are systematically of weaker credit quality across multiple dimensions. These different risk types require different responses.

Implementation: For California — develop HPI stress scenarios (10% and 20% decline) modelling the impact on portfolio LTV distribution and expected delinquency uplift. For New York — increase monitoring

frequency and track modification concentration quarterly. For Florida and Texas — maintain enhanced monitoring given their combination of elevated exposure and above-average delinquency.

Recommendation 5 — Continue Serious Delinquency Migration Surveillance

Business reasoning: Serious delinquency at 0.22% is below the 0.5% threshold. But the COVID-19 forbearance environment suppresses its observed level. As programmes wind down, some borrowers classified as current under relief agreements will return to delinquent status. Tracking the 30→60→90 DPD transition provides the earliest possible warning of credit deterioration otherwise masked by the current environment.

Implementation: Monthly monitoring of delinquency transition rates across the five priority segments. Particular focus on Modified Loan and High Risk Borrower transitions, where progression carries the highest marginal financial impact.

15 PROJECT LIMITATIONS

A rigorous analytical project requires explicit acknowledgement of its boundaries. The following limitations define the conditions under which the findings and recommendations should be interpreted, documented in the spirit of professional analytical integrity.

Limitation	Description	Impact on Analysis
COVID-19 Policy Suppression	Federal forbearance and foreclosure moratoria active during 2020 Q1–Q2 classify stressed borrowers as current and suspend foreclosure proceedings.	Reported delinquency rates are a floor of true stress. Foreclosure near-zero reflects policy, not portfolio health.
Snapshot Model for Dashboard	Power BI KPIs use the latest-snapshot model (1.92M unique loans). Historical transitions and cure rates are not visible in dashboard metrics.	Roll rate and cure rate analysis require the full panel and cannot be performed at the dashboard layer.
No LGD from Data	Insufficient foreclosure exit events occurred during the moratorium to support reliable LGD estimation from actual recovery data.	Loss severity estimates are directional only, relying on industry benchmarks rather than portfolio-specific experience.
Original LTV Only	Only origination LTV available. Post-origination property value changes are not captured.	LTV risk analysis reflects origination leverage. Current LTV may differ materially for seasoned loans depending on local HPI.
DTI as Income Proxy	DTI at origination is the only income-related variable. No current income or employment status is captured.	Payment capacity analysis is limited to origination DTI. COVID-related income loss is not visible in the variable.
Conforming Conventional Scope	Dataset covers Fannie Mae-acquired conforming conventional loans only. FHA, VA, jumbo, and non-agency excluded.	Findings represent the conforming conventional segment and may not generalise to the broader US mortgage market.
No Macroeconomic Variables	No unemployment rates, GDP data, or house price indices in the dataset.	Stress testing requires external macro data supplementation. The analysis cannot project how economic deterioration translates into delinquency uplift.

One intentional analytical exclusion warrants specific explanation: roll rate and cure rate analysis were not developed in the dashboard. These metrics require longitudinal panel data and a monthly observation grain. The latest-snapshot model used for dashboard reporting shows only the current state of each loan, making transition analysis architecturally impossible at the dashboard layer. The full 84.18 million observation panel supports roll rate analysis in SQL, and this represents the highest-priority future analytical enhancement for this project.

16 LESSONS LEARNED

Technical Lessons

Data Architecture Decisions Have Cascading Consequences

The most impactful technical lesson from this project is that the choice of data grain — monthly panel versus latest snapshot — determines what every downstream analysis can and cannot do. This decision needed to be made deliberately and early, because changing it after the analytical layer is built requires rebuilding the entire downstream structure. In hindsight, defining the reporting grain as an explicit architectural decision at the start of the project — rather than discovering the need for a separate snapshot model during Power BI development — would have saved significant rework.

TECHNICAL NOTE *In production credit risk analytics environments, this decision is typically made at the data warehouse design stage: operational reporting tables (current position) and historical analysis tables (full transaction history) are maintained as separate physical structures from the outset, each optimised for its intended use case.*

Column Naming at Import Matters

Loading all 108 columns as Column1 through Column108 required a complete field-mapping exercise before any analysis could begin. In a production environment, this mapping would be applied at the import stage using a configuration file that assigns meaningful column names on ingestion. The lesson: always apply field naming at the earliest possible stage to avoid repeated mapping reference throughout the analytical pipeline.

Pre-Aggregated Summary Tables Are Not Optional at Scale

Running exploratory queries directly against 84 million rows produced unacceptably slow feedback cycles. Creating pre-aggregated summary tables — one for each key analytical dimension — transformed query response times from minutes to seconds and enabled the iterative analytical workflow that good EDA requires. At datasets above approximately 10 million rows, summary table pre-computation is a necessity, not an optimisation.

Business and Credit Risk Lessons

Delinquency Rate Without Exposure Is an Incomplete Risk Metric

The Moderate Risk segment finding — highest delinquent exposure despite not having the highest delinquency rate — is one of the project's most important business lessons. Any risk monitoring system that ranks segments purely by delinquency rate will systematically underweight segments with large exposure and moderate delinquency while overweighting segments with extreme delinquency but small exposure. The combination of rate and exposure is always required for sound portfolio management.

Modification Status Is Not a Proxy Variable — It Is an Independent Signal

Before running the cross-segment analysis on Page 5, a reasonable assumption would have been that modification flags simply identify high-risk borrowers. The cross-segment data definitively refuted this. A Low Risk modified borrower is 59 times more likely to be delinquent than a Low Risk non-modified borrower. The modification flag captures something real and independent that the risk segment variable cannot — the history of financial distress severe enough to require formal intervention.

COVID Context Cannot Be Appended After the Fact

The COVID-19 forbearance environment shaped every metric in this project. Attempting to add that context as a disclaimer at the end of the analysis — rather than embedding it into the interpretation from the beginning — would have produced misleading conclusions. The lesson for all time-period-specific analyses: understand the macroeconomic policy environment of the data before drawing any conclusions, not after.

Dashboard Design Lessons

One Question Per Page Creates Coherence

The discipline of defining one executive business question per dashboard page and refusing to add visuals that did not serve that question was the single most important design decision of the project. Pages that tried to answer two questions became cluttered and lost narrative focus. Pages anchored to a single clear question were immediately comprehensible.

The Reporting Grain Difference Must Be Communicated Proactively

When the SQL analysis produces \$19.97 trillion in exposure and the dashboard shows \$194 billion, any executive who sees both numbers without explanation will immediately question the data quality. Proactive documentation of the methodology difference — explaining both the grain distinction and the business rationale for each approach — is essential for maintaining credibility. The reporting note added to the dashboard and executive report prevented this from becoming a credibility issue.

Diagnostic Pages Are More Valuable Than Ranking Pages

The evolution of Page 4 from 'which states are risky' to 'why are certain states riskier' was the most significant improvement in the dashboard design process. A ranking page answers a question executives can partially answer themselves from their existing knowledge. A diagnostic page answers a question they cannot answer without the analysis — and therefore delivers genuine analytical value.

17 FUTURE IMPROVEMENTS

The analytical framework established in this project provides a solid foundation for several significant enhancements that would substantially deepen its analytical and business value:

Roll Rate and Cure Rate Analysis

Roll rates — the monthly transition probability of a loan moving from one delinquency stage to the next — and cure rates — the probability of a delinquent loan returning to current status — are among the most valuable metrics in mortgage portfolio surveillance. This project's SQL panel (84.18 million observations) contains all the data required. The enhancement requires a self-join on the panel dataset matching each loan's status in month T to its status in month T+1, then calculating transition probabilities for each DPD stage. Incorporating roll rate analysis would enable forecasting of future serious delinquency rates based on current early-stage delinquency momentum.

Probability of Default Modelling

The feature engineering already developed provides the inputs for a formal Probability of Default (PD) model. A logistic regression model trained on the 84.18 million observation panel, predicting 90+ DPD occurrence within a 12-month window, would produce loan-level PD scores that could be incorporated into an IFRS 9-style Expected Credit Loss framework. The AUC of such a model, based on the strong univariate relationships identified in EDA, would likely exceed 0.70 — making it both statistically useful and business-defensible.

Expected Credit Loss (ECL) Framework

Building on PD modelling, a simplified IFRS 9 ECL framework could be developed: $ECL = PD \times EAD \times LGD$. EAD is directly available as current UPB. LGD can be estimated from industry benchmark assumptions or supplemented with actual recovery data from the small number of exit events available. A directional ECL calculation by segment — showing the estimated credit loss embedded in each risk tier — would add significant executive communication value and demonstrate direct IFRS 9 awareness.

Machine Learning Classification

A gradient boosting model (XGBoost or LightGBM) trained on the panel dataset to predict 90+ DPD occurrence would likely produce higher predictive accuracy than a logistic regression baseline, particularly for capturing the non-linear interactions between risk factors that the risk-layering analysis suggests are present. Feature importance analysis from the ML model would also provide empirical validation of the manual risk factor prioritisation used in this project.

Incorporating External Macroeconomic Data

Supplementing the Fannie Mae dataset with FHFA House Price Index data by state and MSA, and unemployment rate data from the Bureau of Labor Statistics, would enable macro-adjusted delinquency analysis and formal stress scenario modelling — the geographic stress testing enhancement identified as a priority in the recommendations.

Time-Series Surveillance Extension

Extending the analysis across additional quarterly datasets — 2020 Q3, Q4, 2021 Q1, and beyond — would transform this from a point-in-time analysis into a genuine longitudinal surveillance framework. The most important analytical question that cannot be answered from 2020 Q1–Q2 data alone is: as forbearance programmes expired, which segments deteriorated most? This longitudinal view would validate the risk prioritisation established in the current project.

18 CONCLUSION

This project began with a question that every mortgage-holding institution must continuously answer: within a large, heterogeneous loan portfolio, which borrowers are most likely to become delinquent, and what should management do about it? Across 84.18 million monthly servicing observations, analysed through PostgreSQL, structured through feature engineering, surfaced through Power BI, and communicated through a formal Executive Credit Risk Report, this project delivered a rigorous, evidence-based answer to that question.

The analytical journey was designed to mirror the actual process of credit risk portfolio analysis in financial institutions. It began with business problem definition and objective setting — establishing what questions needed to be answered and for whom. It moved through data engineering — cleaning, typing, and feature-building a dataset of institutional scale. It progressed through exploratory analysis — systematically interrogating the portfolio across borrower, structural, geographic, and behavioural dimensions. And it concluded with executive communication — translating complex multi-dimensional findings into a six-page dashboard and a formal credit risk report designed for a Chief Risk Officer audience.

The findings were clear. The portfolio is fundamentally healthy — 99.3% of loans are performing, and its borrower quality composition provides structural protection against broad-based deterioration. But within that healthy portfolio, risk is concentrated and identifiable. Modified loans, the strongest behavioural risk signal at 22.8% delinquency, represent the primary monitoring priority. High-risk layered borrowers, purchase loan originations, and New York's multi-dimensional risk profile represent the secondary monitoring landscape.

The portfolio surveillance framework established in Page 6 converts those findings into operational discipline: five monitoring segments, five defined thresholds, five escalation status indicators, and a clear priority ranking for management resource allocation. One segment — Modified Loans — is currently in breach of its threshold. Four segments are within established limits. The overall portfolio risk posture is stable, with one active escalation point.

From a professional development perspective, this project demonstrates the ability to operate across the complete analytics value chain that credit risk employers actually need: understanding the business problem, engineering the data at production scale, conducting rigorous multi-dimensional analysis, designing executive-grade communication products, and translating analytical findings into actionable business recommendations. Each of those capabilities was exercised at institutional data scale — 84 million rows, a Fannie Mae real-world dataset, an industry-standard analytical framework.

Closing statement: *This Mortgage Delinquency Risk Analysis project represents a complete, portfolio-grade demonstration of credit risk analytics applied to real institutional data. It is designed to withstand technical review, business challenge, and interview scrutiny — and to serve as substantive evidence that the analysis behind it was conducted with the rigour, business awareness, and communication discipline expected of a professional credit risk analyst. The combination of institutional data scale, structured methodology, executive-focused communication, and operational surveillance output is what distinguishes this project as a genuine reflection of professional credit risk practice rather than an academic exercise.*

Prepared by

Chegwe Favour

Credit Risk & Data Analyst

Portfolio: <https://chegwefavourportfolioo.netlify.app/>

GitHub: <https://github.com/favouritefavil>

LinkedIn: <http://www.linkedin.com/in/favour-chegwe>